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Tumor immunology

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INTRODUCTION

For as long as we can recollect, immunologists have wanted to believe that they would be the chosen ones to knock the stuffing out of the cancer problem. The fortunes of tumor immunology have fluctuated widely, flavor of the month at one time, a subject fit only for no-hopers at another. It looks as though its time has come at last.

The ability to reject transplants of tissue may be traced back a long way down the evolutionary tree—back even as far as the annelid worms. Long before the studies on the involvement of self-major histocompatibility complex (MHC) in immunological responses, Lewis Thomas suggested that the allograft rejection mechanism represented a means by which the body's cells could be kept under **immunological surveillance** so that altered cells with a neoplastic potential could be identified and summarily eliminated. For this to operate, cancer cells must display some new discriminating surface structure which can be recognized by the immune system.

CHANGES ON THE SURFACE OF TUMOR CELLS (figure 18.1)

Virally controlled antigens

A substantial minority of tumors arise through infection with **oncogenic viruses**, Epstein–Barr virus (EBV) in lymphomas, *human T-cell leukemia virus-1* (HTLV-1) in leukemia and papilloma virus in cervical cancers. After infection, the viruses express genes homologous with cellular oncogenes which encode factors affecting growth and cell division. Failure to control these genes therefore leads to potentially malignant transformation. Virally derived peptides associated with surface MHC on the surface of the tumor cell behave as powerful transplantation antigens which generate haplotype-specific cytotoxic T-cells (Tc). All tumors induced by a given virus should carry the same surface antigen, irrespective of their cellular origin, so that immunization with any one of these tumors would confer resistance to subsequent challenge with the others provided that there were no artful mutations by the virus (Milestone 18.1). Unfortunately, viruses are not innately friendly.

Milestone 18.1—Tumors Can Induce Immune Responses

The first convincing evidence for tumor-associated antigens came from the work of Prehn and Main who demonstrated quite clearly that **chemically induced cancers** can induce immune responses to themselves but not to other tumors produced by the same carcinogen (figure M18.1.1a). Tumors induced by **oncogenic viruses** are different in that processed viral peptides are present on the surface of all neoplastic cells bearing the viral genome so that Tc cells raised to one tumor will cross-react with all others produced by the same virus (figure M18.1.1b).

Dramatic advances were made by Boon and colleagues. First, they showed that random mutagenesis of transplantable tumors, i.e. tumors which can be passaged with-

in a pure mouse strain without provoking rejection, can give rise to mutant progeny with strong transplantation antigens. As a result they could not be grown in syngeneic animals with a normal immune system; accordingly they were referred to as **tum-** variants. Boon's team developed a powerful technology (cf. figure 18.2) which enabled them to use Tc clones specific for the tum- variant to screen cosmid clones for the mutant gene. These two breakthroughs, the recognition that mutation in tumors can generate strong transplantation reactions, and the development of the technique for identifying the relevant antigens with Tc cells, heralded really profound developments in tumor immunology and put it firmly on the map as a key area for cancer research.

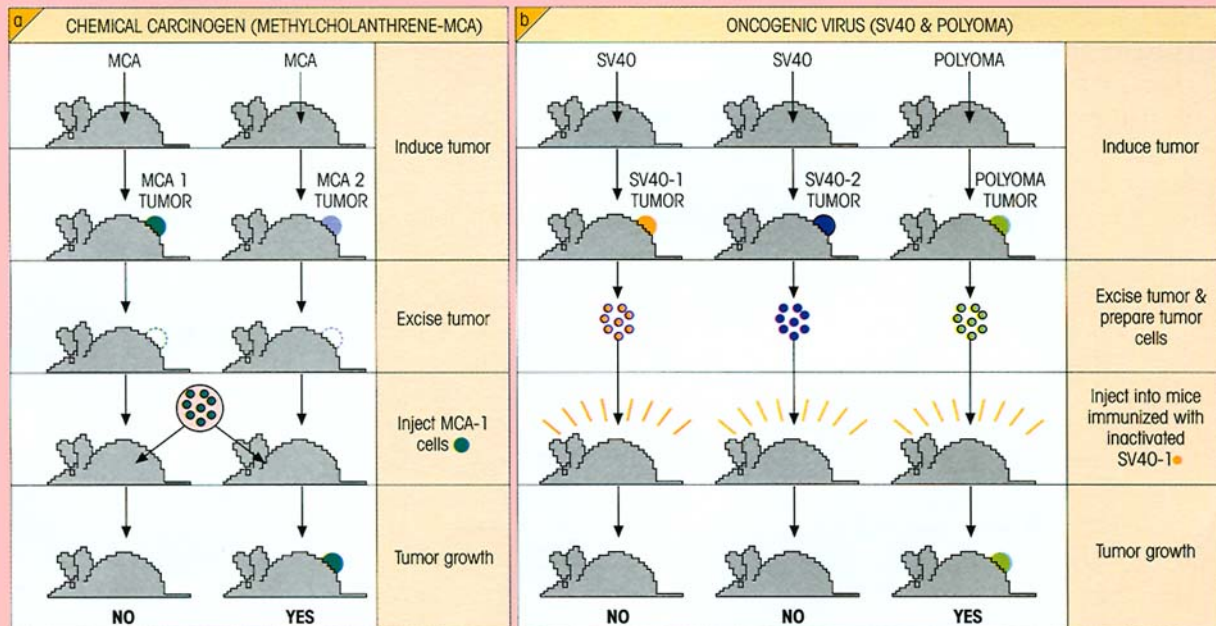


Figure M18.1.1. The specificity of immunity induced by tumors. (a) A chemically induced tumor MCA-1 can induce resistance to an implant of itself but not to a tumor produced in a syngeneic mouse by the same carcinogen. Thus each tumor has an individual antigen, now thought to be a processed mutant endogenous protein complexed with a heat-shock protein. More recent data suggest that, if immunized animals are challenged

with much lower numbers of tumor cells, a greater degree of cross-protection between tumors may be observed, which has been ascribed to a 44 kDa oncofetal antigen, possibly an immature version of a laminin receptor protein. (b) Tumors produced by a given oncogenic virus immunize against tumors produced in syngeneic mice by the same but not other viruses. Thus tumors produced by an oncogenic virus share a common antigen.

Expression of normally silent genes

The dysregulated uncontrolled cell division of the cancer cell creates a milieu in which the products of normally silent genes may be expressed. Sometimes these encode differentiation antigens normally associated with an earlier fetal stage. Thus tumors derived from the same cell type are often found to express such **oncofetal antigens** which are also present on embryonic cells. Examples would be α -fetoprotein in hepatic carcinoma and carcinoembryonic antigen (CEA) in cancer of the intestine. Certain monoclonal antibodies also react with tumors of neural crest origin and fetal melanocytes. Another monoclonal antibody defines the SSEA-1 antigen found on a variety of human tumors and early mouse embryos but absent from adult cells with the exception of human granulocytes and monocytes.

But the exciting quantum leap forward stems from the original observation that cytosolic viral nucleoprotein could provide a target for Tc cells by appearing on the cell surface as a processed peptide associated with MHC class I (cf. p. 94). This established the general principle that the intracellular proteins which are not destined to be positioned in the surface plasma membrane can still signal their presence to T-cells in the outer world by the processed peptide–MHC mecha-

nism. Cytotoxic T-cells specific for tumor cells, obtained from mixed cultures of peripheral blood cells with tumor, can be used to establish the identity of the antigen employing the strategy described in figure 18.2. By something of a *tour de force* a gene encoding a melanoma antigen, MAGE-1, was identified. It belongs to a family of 12 genes, six of which are expressed in a significant proportion of melanomas as well as

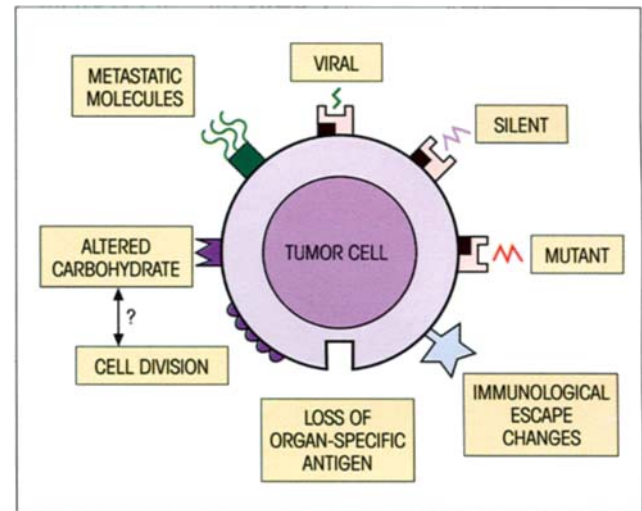


Figure 18.1. Tumor-associated surface changes.

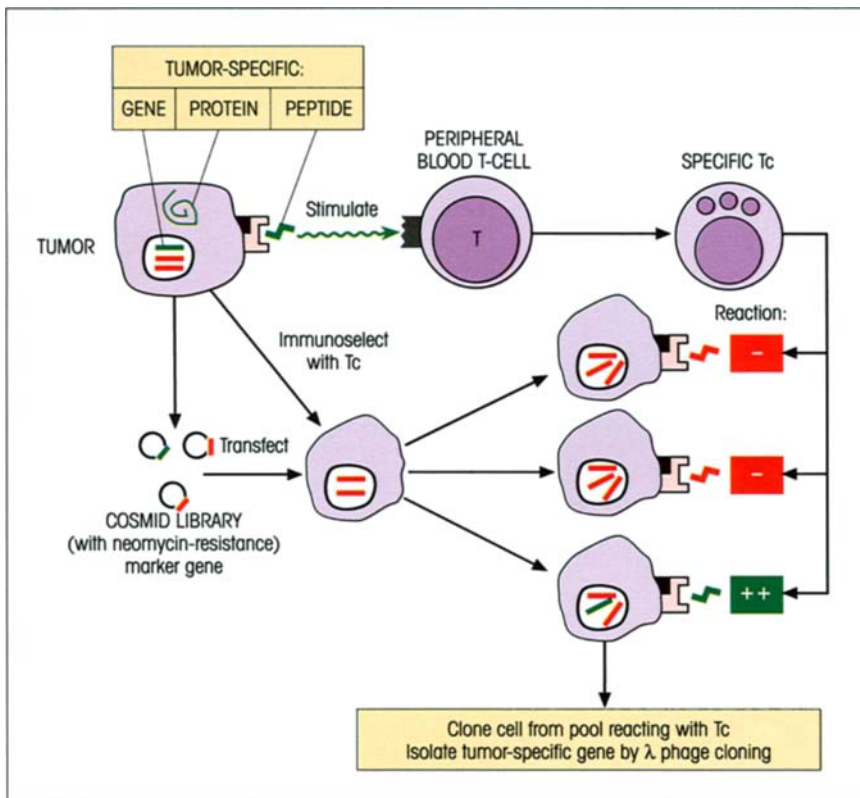


Figure 18.2. Identification of tumor-specific gene using tumor-specific cytotoxic T-cell (Tc) clones derived from mixed tumor–lymphocyte culture. A cosmid library incorporating the tumor DNA is transfected into an antigen-negative cell line derived from the wild-type tumor by immunoselection with the Tc. Small pools of transfected cells are tested against the Tc. A positive pool is cloned by limiting dilution and the tumor-specific gene (*MAGE-1*) cloned from the antigen-positive well(s). (Based on van der Bruggen P. *et al.* (1991) *Science* 254, 1643. Copyright © 1991 by the AAAS.) The original *MAGE-1* belongs to a family of 12 genes. Further melanoma-specific genes, including *MART-1*, *gp100* and *tyrosinase*, have been discovered.

head and neck tumors, nonsmall cell lung cancers and bladder carcinomas. MAGE-1 is *not* expressed in normal tissues except for germ-line cells in testis and gives rise to antigenic T-cell epitopes which, in the light of the absence of class I MHC on the testis cells, must be considered tumor-specific. This exciting research reveals the tumor-specific antigen as an expression of a normally silent gene.

Mutant antigens

The seminal work on tum- mutants (see Milestone 18.1) has persuaded us that single point mutations in oncogenes can account for the large diversity of antigens found on carcinogen-induced tumors. The specific immunity provoked by chemically induced tumors can be elicited by **heat-shock protein 70** (hsp70) and hsp90 isolated from the tumor cells, but their immunogenicity is lost when the associated low molecular weight peptides are removed. These peptides could, however, stimulate the specific CD8 cytotoxic T-cell clones generated by the tumors, and three possible mechanisms have been advanced to account for the enhancement of tumor immune responses by

hsps. First, they can act as 'danger' signals by activating antigen-presenting cells. Second, necrotic tumor cells expressing the hsps can transfer hsp-peptide complexes to host antigen-presenting cells where they can cross-prime cytotoxic CD8 T-cells through the MHC class I endogenous presentation pathway. And last, the hsps may influence the capacity of the tumor cell itself to process and present endogenous mutated and, of course, 'silent' antigens as targets for specific T-cells (figure 18.3).

There is considerable evidence for the production of mutated peptides in human tumors. The gene encoding the p53 cell cycle inhibitor is a hotspot for mutation in cancer and hsp70/p53 peptide complexes have been isolated from human breast tumors. The oncogenic human *ras* genes differ from their normal counterparts by point mutations usually leading to single amino acid substitutions in positions 12, 13 or 61. Such mutations have been recorded in 40% of human colorectal cancers and their preneoplastic lesions, in more than 90% of pancreatic carcinomas, in acute myelogenous leukemia and in preleukemic syndromes. The mutated *ras* peptide can induce proliferative T-cell lines *in vitro*. Tumors induced by UV light produce

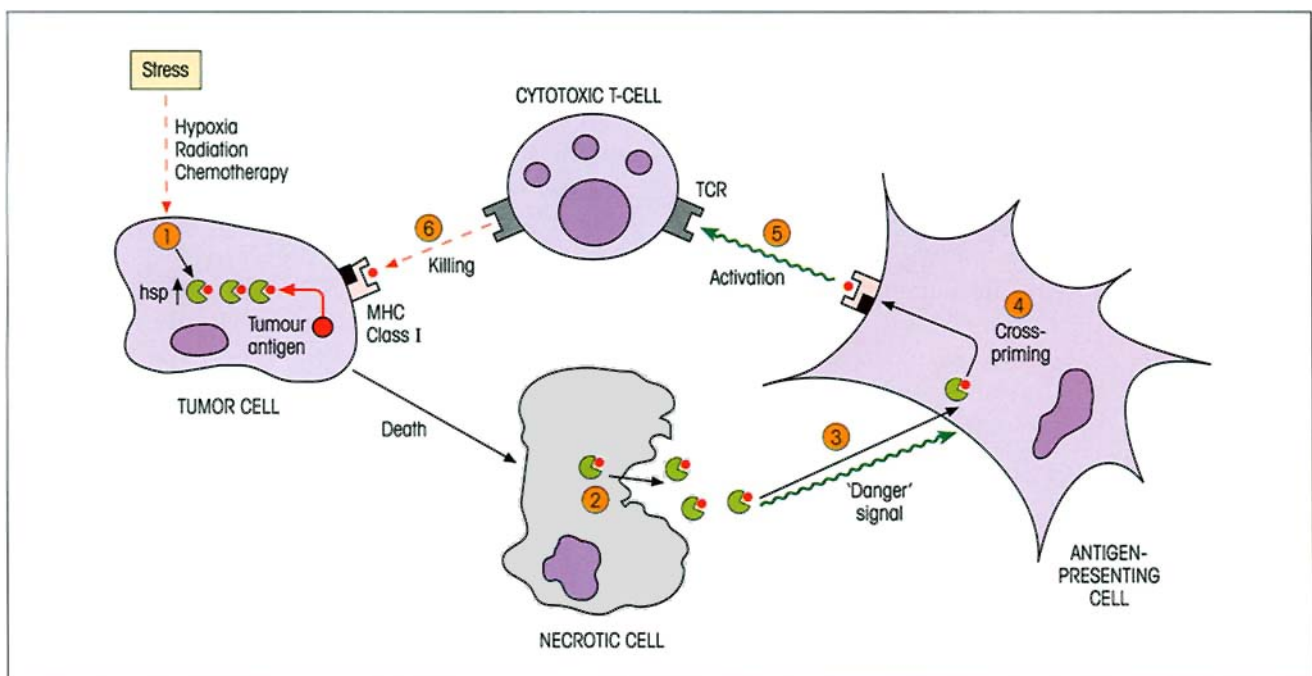


Figure 18.3. The role of heat-shock proteins (hsps) in tumor immunogenicity. (1) Stress factors upregulate hsps which can form complexes with processed tumor antigen and increase surface presentation of antigenic peptide by MHC class I. (2) They can also lead to necrosis and release of hsp-peptide complexes, which (3) can act

as stimulatory danger signals to dendritic antigen-presenting cells and penetrate the cytoplasm, where (4) they can enter the MHC class I processing pathway by so-called cross-priming. (5) CD8 resting T-cells become activated and (6) kill the tumor cells. (Based on Wells A.D. & Malkowsky M. (2000) *Immunology Today* 21, 129.)

individual Tc cells not reacting with other tumors or normal cells and it would be surprising if this were not a further example of tumor-specific mutant peptide antigenicity.

Changes in carbohydrate structure

The chaotic internal control of metabolism within neoplastic cells often leads to the presentation of abnormal surface carbohydrate structures. Sometimes one sees blocked synthesis, e.g. deletion of blood group A. In other cases there may be enhanced synthesis of structures absent in progenitor cells: thus some gastrointestinal cancers express the Lewis Le^a antigen in individuals who are Le(a⁻,b⁻) and others produce extended chains bearing dimeric Le^a or Le(a,b).

Abnormal mucin synthesis can have immunological consequences. Consider the mucins of pancreatic and breast tissue. These consist of a polypeptide core of 20-amino acid tandem repeats with truly abundant O-linked carbohydrate chains. A monoclonal antibody SM-3 directed to the core polypeptide reacts poorly with normal tissue where the epitope is masked by glycosylation, but well with breast and pancreatic carcinomas possessing shorter and fewer O-linked chains. Tc cells specific for tumor mucins are not MHC restricted and the slightly heretical suggestion has been made that the T-cell receptors (TCRs) are binding multivalently to closely spaced SM-3 epitopes on unprocessed mucins; alternatively, and closer to the party line, recognition is by $\gamma\delta$ cells.

Changes on the surface of cycling cells

In some instances it is possible that the changes which occur in the carbohydrate moiety of tumor surface membrane glycoproteins are a natural consequence of cell division. For example, Thomas found that the density of surface sugar determinants cross-reacting with blood group H fell as murine mastocytoma cells moved into the G1 phase of the division cycle, while, reciprocally, group B determinants increased. Surface components binding the lectin, wheat-germ agglutinin are poorly represented on resting T- and B-cells but, within 24 hours of stimulation by lymphocyte polyclonal activators and before DNA synthesis begins, high concentrations of lectin-binding sites appear on the surface.

Unusual events may be observed in active **cells associated with tumor growth** and spread as distinct from the tumors themselves. For example, the 95 kDa surface glycoprotein, F19, is expressed in the **reactive stromal fibroblasts** in more than 90% of carcinomas of

the colon, breast, lung and pancreas, making them a novel target for immunological attack since normal adult tissues and benign epithelial tumors express zero or very low levels. In like vein, the highly sialylated cell surface glycoprotein endosialin (FB5) is present in the newly generated vasculature of a significant proportion of malignant tumors but not in blood vessels of normal tissues.

Molecules related to metastatic potential

Changes in surface carbohydrates can have a dramatic effect on malignancy. For example, colonic cancers expressing sialyl Le^x have a poor prognosis and higher propensity to metastasize. Lung cancer patients whose tumors showed deletion of blood group A had a much worse prognosis than those with continuous A; the finding that patients expressing H/Le^y/Le^b also had a poorer prognosis than antigen-negative subjects is consistent with this observation.

The role of **CD44** (HERMES/Pgp-1) in cell trafficking, based on its interaction with vascular endothelium, has afforded it some prominence in the facilitation of metastatic spread. CD44 occurs in several isoforms with a varying number of exons between the transmembrane and common N-terminus. Normal epithelium expresses the CD44H isoform with hyaluran-binding domains, but lacking the intervening v1–v10 exons; expression of certain of these exons on tumors is indicative of a growth advantage, since they are present with higher frequency on more advanced cancers. Stable transfection of a nonmetastatic tumor with a CD44 cDNA clone encompassing exons v6 and v7 induced the ability to form metastatic tumors—a most striking effect. Further, injection of a monoclonal anti-CD44 v6 prevented the formation of lymph node metastases. Exons v6 and v10 have now been shown to bind blood group H and chondroitin 4-sulfate, respectively, and the latest hypothesis is that these carbohydrates can bind to CD44H on endothelium and thence homotypically to each other so generating a metastatic nidus.

Changes have quite frequently been observed in the expression of class I MHC molecules. For example, oncogenic transformation of cells infected with adenovirus 12 is associated with highly reduced class I as a consequence of very low levels of TAP-1 and -2 mRNA. Mutation frequently leads to diminished or absent class I expression linked in most cases to increased metastatic potential, presumably reflecting decreased vulnerability to T-cells but not NK cells. In breast cancer, for example, around 60% of metastatic tumors lack class I.

SPONTANEOUS IMMUNE RESPONSES TO TUMORS

Immune surveillance against strongly immunogenic tumors

When present, many of these antigens can provoke immune responses in experimental animals which lead to resistance against tumor growth, but they vary tremendously in their efficiency. Powerful antigens associated with tumors induced by oncogenic viruses or ultraviolet light generate strong resistance, while the transplantation antigens on chemically induced tumors (Milestone 18.1) are weaker and somewhat variable; disappointingly, tumors which arise spontaneously in animals produce little or no response. The **immune surveillance theory** would predict that there should be more tumors in individuals whose adaptive immune systems are suppressed. This undoubtedly seems to be the case for **strongly immunogenic tumors**. There is a considerable increase in skin cancer in immunosuppressed patients living in high sunshine regions north of Brisbane and, in general, transplant patients on immunosuppressive drugs are unduly susceptible to skin cancers, largely associated with papilloma virus, and EBV-positive lymphomas. The EBV-related Burkitt's lymphomas crop up with undue frequency in regions infested with malarial infection, known to compromise the efficacy of the immune system. Likewise, the lymphomas which arise in children with T-cell deficiency linked to Wiskott–Aldrich syndrome or ataxia telangiectasia express *EBV* genes; they show unusually restricted expression of EBV latent proteins which are the major potential target epitopes for immune recognition, while cellular adhesion molecules, such as *intercellular adhesion molecule-1* (ICAM-1) and *lymphocyte function-associated molecule-3* (LFA-3), which mediate conjugate formation with Tc cells, cannot be detected on their surface (figure 18.4). Knowing that most normal individuals have highly efficient EBV-specific Tc cells, this must be telling us that only by downregulating appropriate surface molecules can the lymphoma cells escape even the limited T-cell surveillance operating in these patients.

As might be predicted, tumors are positively brimming with various **immunological escape mechanisms** (figure 18.4) and thus they resemble successful infections. Downregulation of HLA class I to make the tumor a less attractive target for cytolytic T-cells is a favorite ploy. We have already referred to this as a common feature of breast cancer metastases, and this is true also of cervical carcinoma where, prognostical-

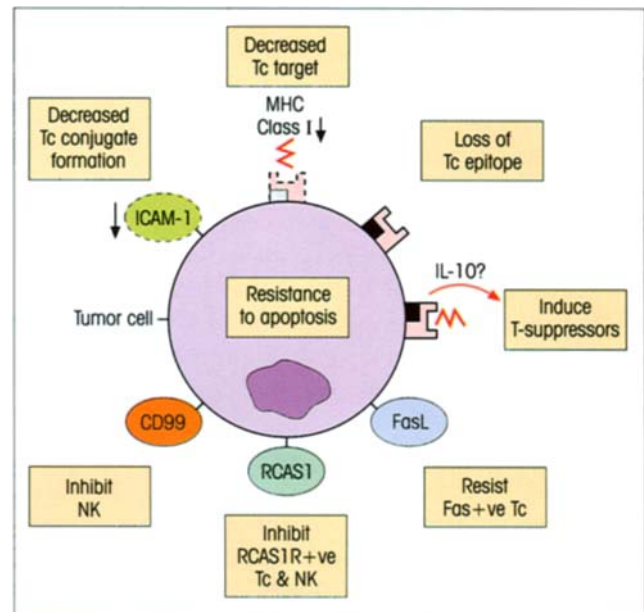


Figure 18.4. Tumor escape mechanisms.

ly, loss of HLA-B44 in premalignant lesions is an indicator of tumor progression. Loss of tumor antigen epitopes represents another escape mechanism and mutations in an oncogenic virus itself can increase its tumorigenic potential. Thus the frequent association of a high-risk variant of human papilloma virus with cervical tumors in HLA-B7 individuals is attributed to the loss of a T-cell epitope which would otherwise generate a protective B7-mediated cytolytic response. Tumors can also decrease their vulnerability to cytotoxic T-cell attack by expression of surface FasL (cf. p. 19) and a growth inhibitory molecule, RCAS1, which react with T-cells bearing their corresponding receptors and stop them in their tracks. It should also be borne in mind that the internal defects in apoptosis which favor the emergence of a tumor clone will, in addition, make them more difficult to kill by cytotoxic T-cells. In a way, it is comforting to draw the conclusion that the existence of these 'Houdini' mechanisms is evidence for a selective pressure being exerted by the adaptive immune system.

Cancers which express neoantigens of **low immunogenicity** do not come creeping out of the woodwork when patients are radically immunosuppressed and, although T-cell responses can often be rescued from tumor-infiltrating lymphocytes or relatively high numbers of tumor-specific CD8 T-cells may be detected by the peptide–HLA tetramer technique in peripheral blood, they may be functionally deficient due perhaps to suppression by local IL-10 and TGF β . Mutation of p53 and its overexpression are very common

events in human cancer and are often associated with the production of antibodies; but while these could prove to have a diagnostic utility, it is most unlikely that they are of benefit to the patient, the current view being that *cell-mediated* responses are crucial to the attack against internal antigens expressed in solid tumors. Reluctantly, one has to accept the view that, with tumors of weak immunogenicity, we are dealing with low-key reactions which clearly play little role in curbing the neoplastic process. That is not to say that these 'weak' antigens cannot be exploited for therapeutic purposes as we shall soon see.

A role for innate immunity?

Perhaps in speaking of immunity to tumors, one too readily thinks only in terms of acquired responses, whereas it is now accepted that innate mechanisms are of significance. Macrophages which often infiltrate a tumor mass can destroy tumor cells in tissue culture through the copious production of reactive oxygen intermediates (ROIs) and tumor necrosis factor (TNF) when activated by a diversity of factors, bacterial lipopolysaccharide, double-stranded RNA, T-cell γ -interferon (IFN γ), and so forth.

There is an uncommon flurry of serious interest in **natural killer (NK)** cells. It is generally accepted that they subserve a function as the earliest cellular effector mechanism against dissemination of blood-borne metastases. Let's look at the evidence. Patients with advanced metastatic disease often have abnormal NK activity and low levels appear to predict subsequent metastases. In experimental animals, removal of NK cells from mice with surgically resected B16 melanoma resulted in uncontrolled metastatic disease and death. Acute ethanol intoxication in rats boosted the number of metastases from an NK-sensitive tumor 10-fold, but had no effect on an NK-resistant cancer, hinting at a possible underlying cause for the association between alcoholism, infectious disease and malignancies. (Those who enjoy the odd Bacchanalian splurge should not be too upset—be comforted by the beneficial effect in heart disease, but no excesses please!) Powerful evidence implicating these cells in protection against cancer is provided by beige mice which congenitally lack NK cells. They die with spontaneous tumors earlier than their nondeficient +/bg littermates, and the incidence of radiation-induced leukemia is reduced by prior injection of cloned isogeneic NK cells which could be suppressing preleukemic cells. Note, however, that tumors induced chemically or with murine leukemia virus were handled normally.

Resting NK cells are spontaneously cytolytic for certain, but by no means all, tumor targets; cells activated by IL-2 and possibly by IL-12 and -18 display a wider lethality. As described earlier in Chapter 4, recognition of the surface structures on the target cell involves various activating and inhibitory receptors, but it is important to re-emphasize that recognition of class I imparts a **negative inactivating** signal to the NK cell. Conversely, this implies that downregulation of MHC class I, which tumors employ as a strategy to escape Tc cells (figure 18.4), would make them **more susceptible to NK attack**. The tumor cells can fight back by expressing CD99, which downregulates NK CD16, and the growth inhibitor RCAS1, which induces apoptosis in NK as well as in Tc cells (figure 18.4). It is not clear whether surface FasL, which can repel attack by the Fas-positive cytolytic T-cells, is also effective against NK cells, but the relative resistance of tumor cells to apoptosis must be innately protective.

Divisions are surfacing in the NK ranks. The NK cells, which remarkably constitute up to 50% of the liver-associated lymphocytes in humans, have a higher level of expression of IL-2 receptor and adhesion molecules such as integrins compared with NK cells in peripheral blood. They are precursors of a subset of activated adherent NK cells (A-NK) which adhere rapidly to solid surfaces under the influence of IL-2 and are distinguished from their nonadherent counterparts by their superiority in entering solid tumors and in prolonging survival following adoptive transfer with IL-2 into animal models of tumor growth or metastasis. The nonadherent NK variety are better at killing antibody-coated cancer cells through antibody-dependent cellular cytotoxicity (ADCC), mediated by their CD16 Fc γ RIII receptor.

Be kind to your NK cells. Really late nights which involve major curtailment of slow-wave sleep lead to drastic falls in NK cells and levels of IL-2, quite apart from bleary eyes.

UNREGULATED DEVELOPMENT GIVES RISE TO LYMPHOPROLIFERATIVE DISORDERS

We should now turn our attention to the manner in which cells involved in immune responses themselves may undergo malignant transformation, giving rise to leukemia, lymphoma or myeloma characterized by uncontrolled proliferation. An obvious example is the subset of adult human T-cell leukemia associated with HTLV-1 (human T-cell leukemia virus type 1). After infection of the T-cell, the viral tax protein, which is constitutively expressed, stimulates transcription of

IL-2, *IL-2R*, etc., leading to vigorous proliferation; however, only if there is a subsequent chromosome abnormality (see below) does malignant transformation take place.

Deregulation of protooncogenes is a characteristic feature of many lymphocytic tumors

The realization that viral oncogenes are almost certainly derived from normal host genes concerned in the regulation of cellular proliferation has led to the identification of many of these so-called protooncogenes. One of them, *c-myc*, appears to be of crucial importance for entry of the lymphocyte, and probably many other cells, from the resting G0 stage to the cell cycle, while shutdown of *c-myc* expression is linked to exit from the cycle and return to G0. Thus deregulation of *c-myc* expression will prevent cells from leaving the cycle and consign them to a fate of continuous replication. This is just what is seen in many of the neoplastic B-lymphoproliferative disorders, where the malignant cells express high levels of *c-myc* protein usually associated with a reciprocal chromosomal translocation involving the *c-myc* locus. For example, Burkitt's lymphoma is a B-cell neoplasia with a relatively high incidence among African children in whom there is an association with the EBV; in most cases studied, the *c-myc* gene, located on chromosome 8 band q24, is joined by a reciprocal translocation event to the μ heavy chain gene on chromosome 14 band q32 (figure 18.5). It is suggested that the normal mechanisms which downregulate *c-myc* can no longer work on the translocated gene and so the cell is held in the cycling mode. Less frequently, *c-myc* translocates to the site of the κ (chromosome 2) or λ (chromosome 22) loci.

Chromosome translocations are common in lymphoproliferative disorders

Most lymphomas and leukemias have visible chromosome abnormalities bound up with translocations to B-cell immunoglobulin or T-cell receptor gene loci but not necessarily involving *c-myc*. A reciprocal translocation between the μ chain gene on chromosome 14 and the *bcl-2* oncogene on chromosome 18 has been identified in almost all follicular B-cell lymphomas, and another between the T-cell gene on chromosome 14 (q11) and another presumed oncogene on chromosome 11 in a T-cell acute lymphoblastic leukemia (TALL).

Lymphomagenesis is a multistep process. The lack of proliferative control engendered by deregulation of *c-myc* and other similar events induced by chromosomal translocations is permissive for the vulnerability to induction of neoplasia, but is not in itself sufficient to bring about malignant transformation. Thus, transgenic mice harboring a *c-myc* gene driven by the μ heavy chain enhancer (E_{μ} -*myc* mice) have hyperplastic expansions of the pre-B-cell population in the bone marrow and spleen during the preneoplastic period and yet do not develop tumors until 6–8 weeks of age; and then they are monoclonal not polyclonal, suggesting that a random second event is required before autonomy is achieved. Indeed, if the E_{μ} -*myc* transgenic mice are now infected with viruses carrying the *v-raf* oncogene, they rapidly develop lymphomas. It is generally thought that, whereas factors like *c-myc* can make a cell competent for mitosis, it is only the additional upregulation of **cell cycle progression genes** like *fos*, *jun* and *myb* or the loss of **cell cycle suppressor gene products** like p53 which allow cell division to occur. Deregulation of two such events may synergize in the process of malignant transformation and the associated unfettered cell proliferation.

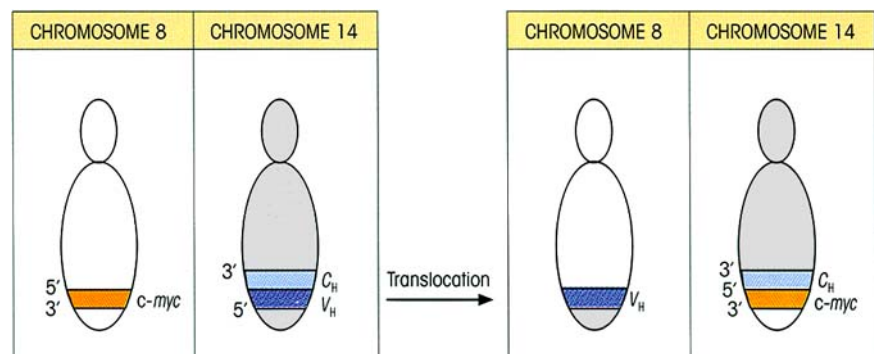


Figure 18.5. Translocation of the *c-myc* gene to the μ chain locus in Burkitt's lymphoma.

Transgenic mice with a susceptibility to the development of cancer may be exploited to supplement the Ames' test for potential carcinogens. Conversely, mice with disrupted *p53* which develop tumors very smartly are ideal for studying treatments which might impede tumorigenesis. The protective effect of caloric restriction on neoplastic change was admirably confirmed in this system.

Different lymphoid malignancies show maturation arrest at characteristic stages in differentiation

Lymphoid cells at almost any stage in their differentiation or maturation may become malignant and proliferate to form a clone of cells which are virtually 'frozen' at a particular developmental stage because of defects in maturation. The malignant cells bear the markers one would expect of normal lymphocytes reaching the stage at which maturation had been arrested. Thus, chronic lymphocytic leukemia cells resemble mature B-cells in expressing surface class II and Ig, albeit of a single idiotype in a given patient. Using monoclonal antibodies directed against the terminal deoxynucleotidyl transferase (see figure 18.7a), class II MHC, Ig and specific antigens on cortical thymocytes, mature T-cells and non-T, non-B acute lymphoblastic leukemia cells, it has been possible to classify the lymphoid malignancies in terms of the phenotype of the equivalent normal cell (figure 18.6).

Susceptibility to malignant transformation is high in lymphocytes at an early stage in ontogeny. If we look at Burkitt's lymphoma, the EBV-induced translocation of the *c-myc* to bring it under control of the *IgH* gene complex is most likely to occur at the pro-B-cell stage, since the chromatin structure of the Ig locus opens up for transcription as signaled by the appearance of sterile C_{μ} transcripts. Furthermore, the cell is likely to escape immunological recognition because, in its undifferentiated resting form, it has downregulated its EBV-encoded antigens and MHC class I polymorphic specificities, and the adhesion molecules LFA-1, LFA-3 and ICAM-1, as mentioned above.

At one time it was thought that maturation arrest occurred at the stage when the cell first became malignant, but we now know that the tumor cells can be forced into differentiation by agents such as phorbol myristate acetate, and the current view is that cells may undergo a few differentiation steps after malignant transformation before coming to a halt. The demonstration of a myeloma protein idiotype on the cytoplasmic μ chains of pre-B-cells in the same patient certainly favors the idea that the malignant event had occurred

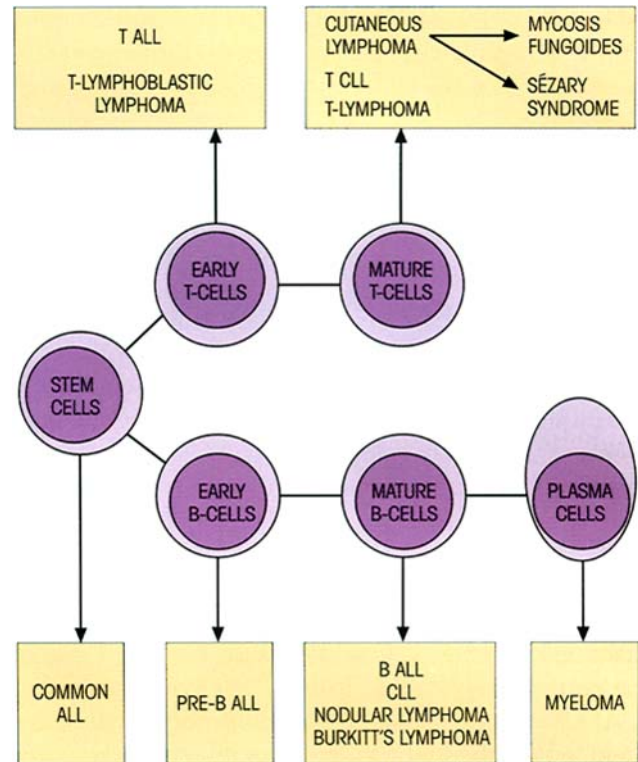


Figure 18.6. Cellular phenotype of human lymphoid malignancies. ALL, acute lymphoblastic leukemia; CLL, chronic lymphocytic leukemia. (After Greaves M.F. & Janossy G, personal communication.)

in a pre-B-cell whose progeny formed the plasma cell tumor. However, an alternative explanation could be transfection of normal pre-B-cells by an oncogene complex from the myeloma cells, possibly through a viral vector. With the exciting discovery of retroviruses associated with certain human T-cell leukemias, this is an interesting possibility.

Immunohistological diagnosis of lymphoid neoplasias

With the availability of a range of monoclonal antibodies and improvements in immuno-enzymatic technology, great strides have been made in exploiting, for diagnostic purposes, the fact that malignant lymphoid cells display the markers of the normal lymphocytes which are their counterparts.

Leukemias

This point can be made rather well if one looks at the markers used to distinguish between the various types of leukemia (table 18.1). Whereas T ALL and B ALL

Table 18.1. Classification of lymphocytic leukemia by immunoenzymatic staining.

Lymphocyte marker	Common ALL	Pre-B ALL	B-cell ALL	T-cell ALL	Chronic lymphocytic leukemia
*CALLA (CD10)	+	+	–	–	–
Cytoplasmic μ	–	+	–	–	–
Surface μ	–	–	+	–	+
Surface κ or λ	–	–	+	–	+
Pan-B	–	+	+	–	+
TdT	+	+	–	+	–
CD5	–	–	–	+	+
CD2	–	–	–	+	–
HLA-DR	+	+	+	–	+

*Antigen-specific for lymphoid precursor cells and pre-B-cells.

cases have a poor prognosis, the patients positive for the common acute lymphoblastic leukemia antigen (CALLA; figure 18.7b), which includes most childhood leukemias, belong to a prognostically favorable group, many of whom are curable with standard therapeutic combinations of vincristine, prednisolone and L-asparaginase. Bone marrow transplantation may help in the management of patients with recurrent ALL provided that a remission can first be achieved. Expression of the lymphoid lineage markers CD2 and CD19 on leukemic blasts is an indication of good prognosis in adults with the disease.

Chronic lymphocytic leukemia is uncommon in people under 50 years and is usually relatively benign, although the 10–20% of patients with a circulating monoclonal Ig have a bad prognosis. Excessive numbers of CLL small lymphocytes are found in the blood (figure 18.7c–e) and, being derived from a single clone, they can be stained only with anti- κ or anti- λ (table 18.1). Their weak expression of CD5 strongly suggests that they may be derived from the equivalent of the B-1 cell population, especially since they can be encouraged to make the IgM polyspecific autoantibodies typical of this subset, if pushed by phorbol ester stimulation.

Lymphomas

The extensive use of markers has greatly helped in the diagnosis of non-Hodgkin's lymphomas. In the first place, the sometimes difficult distinction between a lymphoproliferative condition and carcinoma can be made with ease by using monoclonal antibodies to the leukocyte common antigen (CD45), which will react

with all lymphoid cells whether in paraffin or cryostat sections, and antibodies to cytokeratin which recognize most carcinomas (figure 18.7f). Second, the cell of origin of the lymphoma can be ascertained by panels of monoclonals which differentiate the cellular elements which form normal lymphoid tissue.

The majority of non-Hodgkin's lymphomas are of B-cell origin and the feature which gives the game away to the diagnostic immunohistologist is the synthesis of monotypic Ig, i.e. of one light chain only (figure 18.7g); in contrast, the population of cells at a site of reactive B-cell hyperplasia will stain for both κ and λ chains (figure 18.7h).

Follicle center cell lymphomas (figure 18.7g) imitating the reactive germinal center account for over 50% of the B-lymphomas. They exhibit monotypic surface Ig, and the larger centrocytes and centroblasts which make up two-thirds of the cases contain cytoplasmic Ig. They stain for MHC class II and weakly for CALLA. Morphologically similar cells make up tumors variously labeled as 'mantle zone lymphoma' or 'small cleaved cell lymphoma' but differ from follicle center cells in positive surface staining for IgM and IgD and CD5, and negativity for CALLA. Burkitt's lymphoma lymphoblastoid cells (figure 18.7i) exhibit the common ALL antigen and surface IgM.

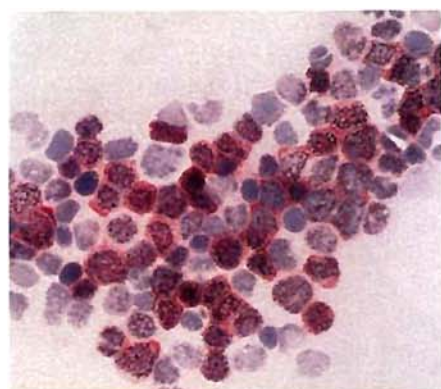
The overall prognosis for patients with non-Hodgkin's lymphoma is poor, even though improved by combined chemotherapy. Transplanted patients are 35 times more likely to develop lymphoma than normals, and there are indications that this cannot necessarily be attributed to the long-term immunosuppression.

Hodgkin's disease attacks the gross architecture of lymphoid tissue and is characterized by the binucleate giant cells known as Reed–Sternberg cells (figure 18.7j) which appear to have a germinal center B-cell lineage. Therapy depends upon the stage of the disease; patients with disease localized to lymphoid tissue above the diaphragm respond well to radiotherapy, while those with more widespread disease are treated more aggressively.

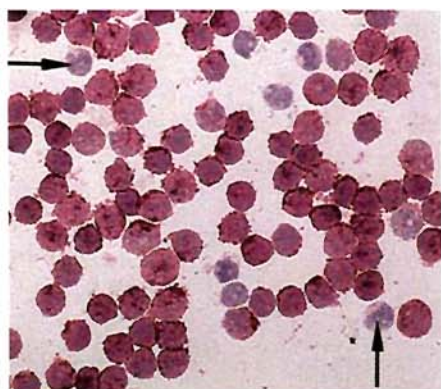
Plasma cell dyscrasias

Multiple myeloma

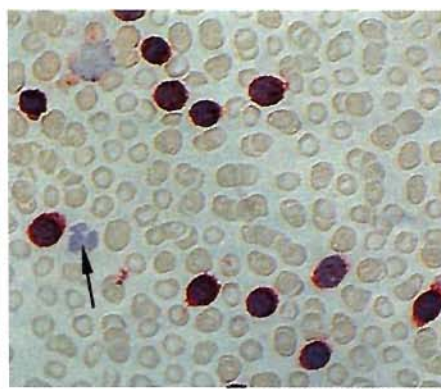
This is defined as a malignant proliferation of a clone of plasma cells in the bone marrow secreting a monoclonal Ig. The myeloma or 'M' component in serum is recognized as a tight band on paper electrophoresis (figure 18.8) as all molecules in the clone are of course identical and have the same mobility. Since Ig-



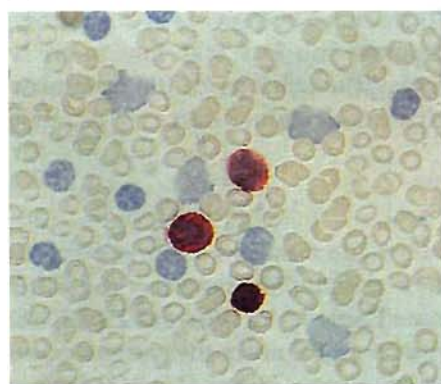
(a)



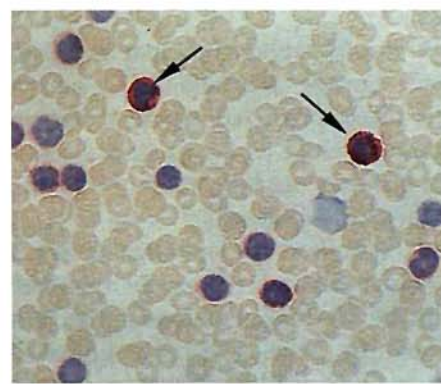
(b)



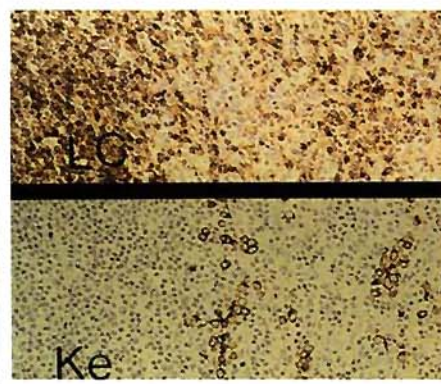
(c)



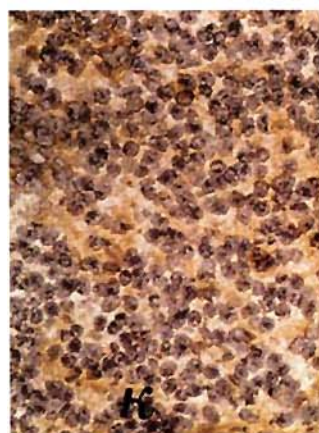
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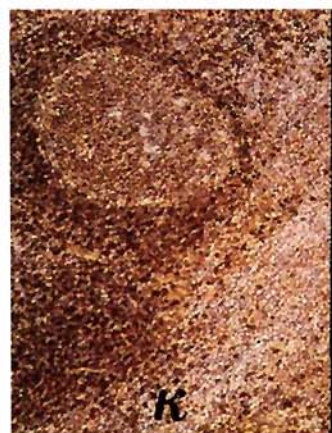
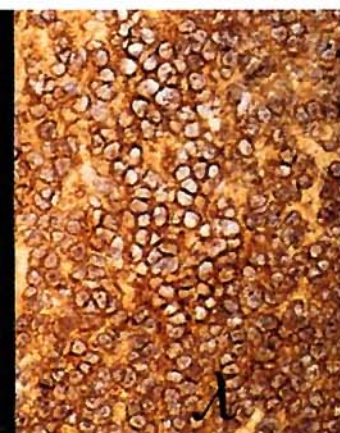
(e)



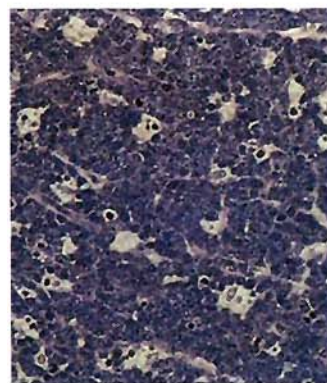
(f)



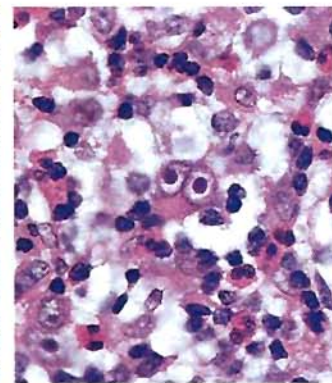
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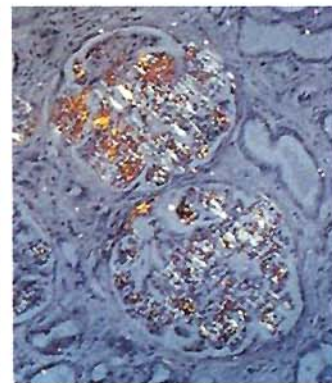
(h)



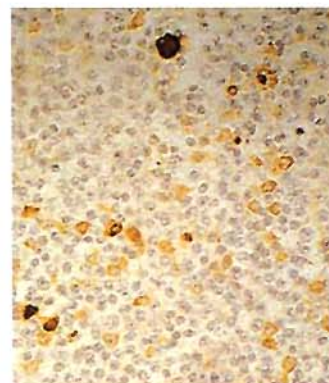
(i)



(j)



(k)



(l)

Figure 18.7. (Opposite) Immunodiagnosis of lymphoproliferative disorders. (a) Cytocentrifuged blast cells from a case of acute lymphoblastic leukemia stained by anti-terminal deoxynucleotidyl transferase (TdT) using an immuno-alkaline phosphatase method (cells treated first with mouse monoclonal anti-TdT, then anti-mouse Ig and finally with an immune complex of mouse anti-alkaline phosphatase + alkaline phosphatase before developing the enzymatic reddish-purple color reaction). Many strongly stained blast cells are seen together with unlabeled normal marrow cells. (b) Immuno-alkaline phosphatase staining of bone marrow cells from a case of acute lymphoblastic leukemia, using monoclonal antibody specific for the common acute lymphoblastic leukemia antigen (anti-CALLA; antibody J5). The majority of cells are strongly labeled. Two nonreactive cells are indicated by arrows. (c, d, e) Immuno-alkaline phosphatase labeling of blood smears from a case of chronic lymphocytic leukemia with three monoclonal antibodies (anti-HLA-DR, anti-CD3 antigen and anti-CD1 antigen): (c) HLA-DR antigen is present on all the leukemic cells seen, but absent from a polymorph (arrowed); (d) three normal T-cells are labeled for the CD3 antigen, but the leukemic cells are negative; (e) CD1 antigen is strongly expressed on two normal lymphocytes (arrowed), but also weakly expressed on the chronic lymphocytic leukemia (CLL) cells. This pattern is typical of CLL. (f) A case of gastric carcinoma (stained at the bottom using anticytokeratin, Ke) with a heavy lymphocytic infiltrate (top, stained with antileukocyte common antigen, LC). (g) Diffuse follicle center type B-cell lymphoma showing λ light chain restriction; compare with (h) a reactive lymph node staining for both κ and λ light chains. (i) Burkitt's lymphoma showing 'starry sky' appearance. (j) Hodgkin's disease showing mixed cellularity and characteristic binucleate Reed-Sternberg cell with massive prominent nucleoli in the center of the figure. (k) Birefringent amyloid deposits in kidney glomeruli visualized by Congo Red staining under polarized light. (l) A case of malignant lymphoma associated with macroglobulinemia, showing lymphoplasmacytoid cells stained by the brown immunoperoxidase reaction for cytoplasmic IgM. ((a)–(e) Very kindly provided by Professor D. Mason, and (f)–(l) by Professor P. Isaacson.)

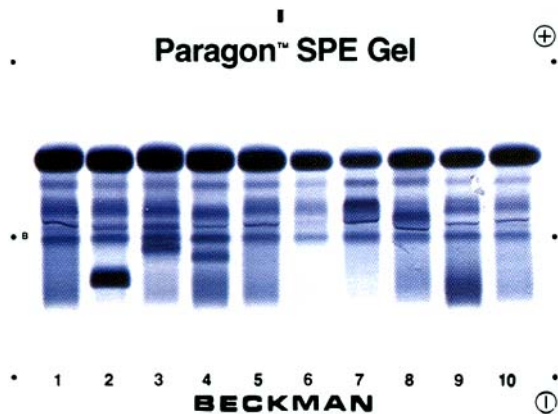


Figure 18.8. Myeloma paraprotein demonstrated by gel electrophoresis of serum. Lane 1, normal; lane 2, γ -paraprotein; lane 3, near β -paraprotein; lane 4, fibrinogen band in the γ region of a plasma sample; lane 5, normal serum; lane 6, immunoglobulin deficiency (low γ); lane 7, nephrotic syndrome (raised α_2 -macroglobulin, low albumin and Igs); lane 8, hemolysed sample (raised hemoglobin/haptoglobin in α_2 region); lane 9, polyclonal increase in Igs (e.g. infection, autoimmune disease); lane 10, normal serum. (Gel kindly provided by Mr A. Heys.)

secreting cells produce an excess of light chains, free light chains are present in the plasma of multiple myeloma patients, can be recognized in the urine as Bence-Jones protein (cf. figure 16.23) and give rise to amyloid deposits (see below). The characteristic 'punched out' osteolytic lesions in bones are thought to be due to the release of osteoclastic factors such as IL-6 by the abnormal plasma cells in the marrow. If untreated, the disease is rapidly progressive. With chemotherapy, the mean survival time from diagnosis is now about 5 years.

'M' bands have been found in the sera of a number of individuals who have no clinical signs of myeloma; the comparative rarity with which invasive multiple myeloma develops in these people and the constant level of the monoclonal protein over a period of years suggest the presence of benign tumors of the lymphocyte-plasma cell series.

Amyloid. Between 10% and 20% of patients with myeloma develop widespread amyloid deposits which contain the variable region of the myeloma light chain. Being identical, the variable region fragments polymerize and form the characteristic amyloid fibrils which are recognizable by their birefringence on staining with Congo Red (figure 18.7k). Other components in amyloid have not yet been characterized. The fibrils are relatively resistant to digestion and accumulate in the ground substance of connective tissue where they can lead to pathological changes in the kidneys, heart and brain. Amyloid can also be formed secondarily to chronic inflammatory conditions such as rheumatoid arthritis and familial Mediterranean fever, but in this case involves the polymerization of a unique substance, Amyloid A (AA), a protein derived from the N-terminal part of a serum precursor (SAA) of molecular weight 90 kDa. SAA behaves as an acute phase protein in that its concentration increases rapidly in response to tissue injury or inflammation. Levels rise with age and the minority of individuals with high values are those most likely to develop amyloid.

Waldenström's macroglobulinemia

This disorder is produced by the unregulated proliferation of cells of an intermediate appearance called lymphoplasmacytoid cells which secrete a monoclonal IgM, the Waldenström macroglobulin (figure 18.7l). Remarkably, many of the monoclonal proteins have autoantibody activity, anti-DNA, anti-IgG (rheumatoid factor), and so on. It has been suggested that, like the CLL cells, they are of the B-1 lineage which secrete 'natural' antibodies (cf. p. 236). Since the IgM is se-

creted in large amounts and is confined to the intravascular compartment, there is a marked rise in serum viscosity, the consequences of which can be temporarily mitigated by vigorous plasmapheresis. The disease runs a fairly benign course and the prognosis is quite good, although the appearance of lymphoplasmacytoid tumor cells in the blood is an ominous sign.

Heavy chain disease

Heavy chain disease is a rare condition in which quantities of abnormal heavy chains are excreted in the urine— γ chains in association with malignant lymphoma and α chains in cases of abdominal lymphoma with diffuse lymphoplasmacytic infiltration of the small intestine. The amino acid sequences of the N-terminal regions of these heavy chains are normal, but they have a deletion extending from part of the variable domain through most of the C_H1 region so that they lack the structure required to form cross-links to the light chains. One idea is that the defect arises through faulty coupling of V- and C-region genes (cf. p. 42).

Immunodeficiency secondary to lymphoproliferative disorders

Immunodeficiency is a common feature in patients with lymphoid malignancies. The reasons for this are still obscure, but it seems as though the malignant cells interfere with the development of the corresponding normal cells, almost as though they were producing some cell-specific chalone (inhibitor) or transfecting suppressor factor. Thus, in multiple myeloma, the levels of normal B-cells and of nonmyeloma Ig may be grossly depressed and the patients may be susceptible to infection with pyogenic bacteria.

APPROACHES TO CANCER IMMUNOTHERAPY

Although immune surveillance seems to operate only against strongly immunogenic tumors, the exciting new information on the antigenicity of mutant and previously silent proteins, and the changes in carbohydrate structures, should be of tremendous encouragement to any red-blooded investigator with an eye to develop new immunotherapies for cancer which exploit the already impressive range of antigens currently available (table 18.2).

On one point all are agreed: if immunotherapy is to succeed, it is essential that the tumor load should first be reduced by surgery, irradiation or chemotherapy,

Table 18.2. Potential tumor antigens for immunotherapy. (Reproduced with permission from Fong L. & Engleman E.G. (2000) Dendritic cells in cancer immunotherapy. *Annual Review of Immunology* 18, 245.)

Antigen	Malignancy
Tumour specific	
Immunoglobulin V-region	B-cell non-Hodgkin's lymphoma, multiple myeloma
TCR V-region	T-cell non-Hodgkin's lymphoma
Mutant p21/ras	Pancreatic, colon, lung cancer
Mutany p53	Colorectal, lung, bladder, head and neck cancer
Developmental	
p210/bcr-abl fusion product	Chronic myelogenous leukemia, acute lymphoblastic leukemia
MART-1/Melan A	Melanoma
MAGE-1, MAGE-3	Melanoma, colorectal, lung, gastric cancers
GAGE family	Melanoma
Telomerase	Various
Viral	
Human papilloma virus	Cervical, penile cancer
Epstein-Barr virus	Burkitt's lymphoma, nasopharyngeal carcinoma, post-transplant lymphoproliferative disorders
Tissue specific	
Tyrosinase	Melanoma
gp100	Melanoma
Prostatic acid phosphatase	Prostate cancer
Prostate-specific antigen	Prostate cancer
Prostate-specific membrane antigen	Prostate cancer
Thyroglobulin	Thyroid cancer
α -fetoprotein	Liver cancer
Overexpressed	
Her-2/ <i>neu</i>	Breast and lung cancers
Carcinoembryonic antigen	Colorectal, lung, breast cancer
Muc-1	Colorectal, pancreatic, ovarian, lung cancer

since not only is it unreasonable to expect the immune system to cope with a large tumor mass, but considerable amounts of antigen released by shedding would tend to prevent the generation of any significant response in some cases due to the stimulation of T-suppressors. This leaves the small secondary deposits as the proper target for immunotherapy.

Antigen-independent cytokine therapy

Sporadic successes have greeted the use of components of the immune system for antigen-independent therapy.

Interleukin treatment

On activation by IL-2 or IL-12, NK cells are capable of killing a variety of fresh tumor cells *in vitro* and, on the basis of studies on mice with mammary glands carrying the HER-2/*neu* oncogene, it would not be unreasonable to conduct a trial of systematic IL-12 treatment in cancer patients with minimum residual disease in an attempt to prevent recurrence and to inhibit incipient metastases.

Interferon therapy

In trials using IFN α and IFN β , a 10–15% objective response rate was seen in patients with renal carcinoma, melanoma and myeloma, an approximate 20% response rate among patients with Kaposi's sarcoma, about 40% positive responders in patients with various lymphomas and a remarkable response rate of 80–90% among patients with hairy cell leukemia and mycosis fungoides.

With regard to the mechanisms of the antitumor effects, in certain tumors IFNs may serve primarily as antiproliferative agents; in others, the activation of NK cells and macrophages may be important, while augmenting the expression of class I MHC molecules may make the tumors more susceptible to control by immune effector mechanisms. In some circumstances the antiviral effect could be contributory.

For diseases like renal cell cancer and hairy cell leukemia, IFNs have induced responses in a significantly higher proportion of patients than have conventional therapies. However, in the wider setting, most investigators consider that their role will be in combination therapy, e.g. with active immunotherapy or with various chemotherapeutic agents where synergistic action has been observed in murine tumor systems. IFN α and β synergize with IFN γ and the latter synergizes with TNF. IFN α acts as a radiation sensitizer and its ability to increase the expression of estrogen receptors on cultured breast cancer cells suggests the possibility of combining IFN with anti-estrogens in this disease.

Colony-stimulating factors

Normal cell development proceeds from an immature stem cell with the capacity for unlimited self-renewal, through committed progenitors, to the final lineage-specific differentiated cells with little or no potential for self-renewal. Therapy aimed at inducing tumor cell differentiation is founded on the idea that the induction of cell maturation decreases and possibly abro-

gates the capacity of the malignant clone to divide. Along these lines, GM-CSF has been shown to enhance the differentiation, decrease the self-renewal capacity and suppress the leukemogenicity of murine myeloid leukemias. Recombinant human products are now undergoing trials.

It is over 100 years since the physician Coley gave his name to the mixture of microbial products termed **Coley's toxin**. This concoction certainly livens up the innate immune system and does produce remission in a minority of patients. The suggestion has been made that these beneficial effects are due to the release of TNF since the vascular endothelium of tumors is unduly susceptible to damage by this cytokine and hemorrhagic necrosis is readily induced. It is questionable whether the critical levels of TNF are reached in the human since these would be very toxic, although one study involving perfusion of an isolated limb with TNF, IFN γ and melphalan provoked lesions in the tumor endothelium without affecting the normal vasculature. Opinion is coming round to the view that the Coley phenomenon may be linked more to boosting a pre-existing weak antitumor immunity.

Exploitation of cell-mediated immune responses

The current dogma is that T-cells rather than antibodies are capable of savaging solid tumors, particularly those expressing processed intracellular antigens on their surface, and, since the majority are MHC class II negative, it looks as though we are aiming at essentially CD8 cytotoxic T-cell responses, although CD4 T-cells can be involved in protective reactions against tumor-associated vasculature.

Virally induced tumors

Based on the not unreasonable belief that certain forms of cancer (e.g. lymphoma) are caused by oncogenic viruses, attempts are being made to isolate the virus and prepare a suitable vaccine from it. In fact, large-scale protection of chickens against the development of Marek's disease lymphoma has been successfully achieved by vaccination with another herpes virus native to turkeys. In human Burkitt's lymphoma, work is in progress to develop a vaccine to exploit the ability of Tc cells to target **EBV-related antigens** on the cells of all Burkitt's tumors. It may be an advantage to treat the patient at the same time with cytokines to upregulate the expression of ICAM-1, LFA-3 and possibly of the virus itself.

Immunization with whole tumor cells

This has the advantage that we do not necessarily have to know the identity of the antigen concerned. The disadvantage is that the majority of tumors are weakly immunogenic, and do not present antigen effectively and so cannot overcome the **barrier to activation of resting T-cells**. Remember, the surface MHC–peptide complex on its own is not enough; costimulation with molecules such as B7.1 and B7.2 and possibly certain cytokines is required to push the G0 T-cell into active proliferation and differentiation. Once we get to this stage, however, the **activated T-cell no longer requires the accessory costimulation** to react with its target, for which it has a greatly increased avidity due to upregulation of accessory binding molecules such as CD2 and LFA-1 (cf. p. 165; figure 18.9). The system works! Vaccination with B7-transfected murine melanoma generated CD8⁺ cytolytic effectors which protected against subsequent tumor challenge; in other words, transfection enabled the melanoma cells to present their own antigens efficiently, while the untransfected cells were vulnerable targets for the cytotoxic T-cells so produced. A further telling observation was that an irradiated nonimmunogenic melanoma line which had been transfected with a retroviral vector carrying the GM-CSF gene stimulated potent and specific antitumor immunity, almost certainly by enhancing the differentiation and activation of host antigen-presenting cells. A less sophisticated but more convenient approach ultimately utilizing similar mechanisms involves the administration of the

irradiated melanoma cells together with BCG which, by generating a plethora of inflammatory cytokines, increases the efficiency of presentation of tumor antigens derived from necrotic cells. In a large-scale study of over 1500 patients, 26% of vaccinees were alive at 5 years compared with only 6% of those treated with the best available conventional therapy. It would be exciting to suppose that in the future we might expose a tumor surgically and then transfect it *in situ* by firing gold particles (cf. p. 302) bearing appropriate gene constructs such as B7, IFN γ (to upregulate MHC class I and II), GM-CSF, IL-2, and so on (figure 18.9). There is a real risk of inducing autoimmune responses to cryptic epitopes (cf. p. 407) shared with other normal tissues which the prudent investigator will not overlook.

Therapy with subunit vaccines

The variety of potential protein targets so far identified (table 18.2) has spawned a considerable investment in clinical therapeutic trials using peptides as vaccines. Because of the pioneering work in characterizing melanoma-specific antigens, this tumor has been the focus of numerous studies which exploit to the full the academic background to modern immunology. Encouraging results in terms of clinical benefit, linked to the generation of cytolytic T-cells (CTLs), have been obtained following vaccination with peptides complexed with heat-shock proteins or modified at class I anchor residues to improve MHC binding. The inclusion of accessory factors, such as IL-2 or GM-CSF, and **CTLA-4 blockade** can be crucial for success. Poten-

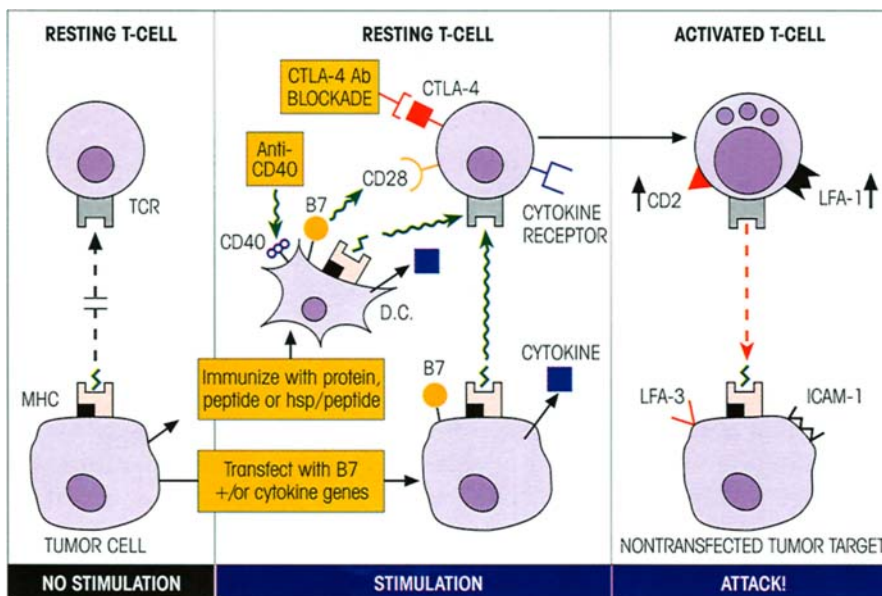


Figure 18.9. Immunotherapy by transfection with costimulatory molecules. The tumor can only stimulate the resting T-cell with the costimulatory help of B7-1 and -2 and/or cytokines such as GM-CSF, γ -interferon and various interleukins, IL-2, -4 and -7. CTLA-4 blockade enhances immunogenicity. Alternatively, the T-cell can be stimulated directly by tumor antigens presented by dendritic cells (DCs) which can themselves be activated by cross-linking their surface CD40 with antibody (see below). Once activated, the T-cell with upregulated accessory molecules can now attack the original tumor lacking costimulators.

tially tolerogenic peptide vaccines can be converted into strong primers for CTL responses by triggering CD40 with a cross-linking antibody which can substitute for T-cell help in the direct activation of CTLs (figure 18.10). Anti-CD40 treatment alone was also shown to partially protect mice bearing *CD40-negative* lymphoma cells, an effect attributed to the activation of endogenous dendritic antigen-presenting cells (cf. figure 18.9).

The unique **idiotype** on monoclonal B-cell tumors is a mouth-watering target for immunotherapy. Although one might expect the unique sequences in the hypervariable CDR loops to be the obvious candidates, one study has revealed that peptides derived from the **immunoglobulin framework** function as commonly expressed CTL epitopes. The advantage here is that one does not have to synthesize an individual idiotype peptide for each patient, while mutations in the idiotypic determinant, which would allow the tumor to escape from the immune response, are far less likely to occur with a framework peptide. Whether this approach will prove ultimately preferable to those employing scFv vaccines cannot be predicted at present. A DNA construct of a tumor scFv fused to a pathogen-derived sequence from tetanus toxin gave excellent protective immunity against a mouse myeloma; there was no surface Ig on the tumor cells and protection must have been afforded through targeting

of processed idiotype complexed with MHC by CD4 and CD8 cells.

Anticipating a more widespread use of naked DNA for immunotherapy, immunogenicity has been enhanced by making the vaccine **self-replicating** using a construct with a gene encoding RNA replicase from Semliki Forest virus. The improved efficacy correlated with caspase-dependent apoptotic death of transfected cells which facilitated their uptake by dendritic cells. Concurrent IL-12 proved to be a potent accessory adjuvant. These alphavirus replicon-based RNA vaccines are noninfectious and reversion cannot occur because the constructs lack structural protein genes; further, since they do not go through a DNA intermediate, there is little chance of genomic integration.

The sheer power of the **dendritic antigen-presenting cell (DC)** for the initiation of T-cell responses has been the focus of an ever-burgeoning series of immunotherapeutic strategies which have elicited tumor-specific protective immune responses via injection of isolated DC loaded with tumor lysates or tumor antigens or peptides derived from them. Considerable success has been achieved in animal models and increasingly with human patients (figure 18.11). The copious numbers of DCs needed for each patient's individual therapy are obtained by expansion of CD34-positive precursors in bone marrow by culture with GM-CSF, IL-4 and TNF, and sometimes with extra goodies such as stem cell factor (SCF) and Fms-like tyrosine kinase 3 (Flt3)-ligand. CD14-positive monocytes from peripheral blood are easier to access, and generate DC in the presence of GM-CSF plus IL-4; however, they need additional maturation with TNF α which increases cost and the chance of bacterial contamination. Another approach is to expand the DCs *in vivo* by administration of Flt3-ligand. The circulating blood DCs increase in number 10–30-fold and can be harvested by leukaphoresis.

Some general points may be made. First, where peptides are used to load the DC, sequences which bind strongly to a given MHC class I haplotype must be identified; sequences will vary between patients with different haplotypes and they may not include potential CD4 helper epitopes. Recombinant proteins will overcome most of these difficulties, and a mixture should be even better since it should recruit more CTLs and be more able to 'ride out' any new tumor antigen mutations. However, proteins taken up by DCs are relatively inefficient at 'cross-priming' CD8 CTLs through the class I processing pathway, although several tactics are being explored to circumvent this problem: they include conjugation to an HIV-tat 'transporter' peptide which increases class I presentation

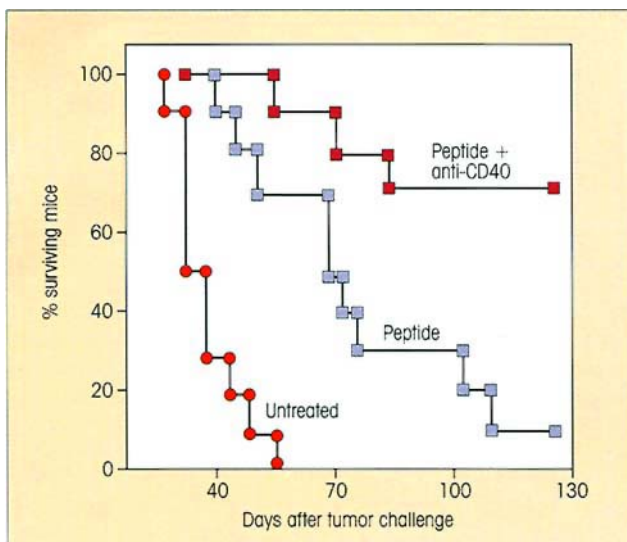


Figure 18.10. CD40 ligation enhances the protective effect of a peptide vaccine against a pre-existing tumor. Six days after injection of human papilloma virus-16 (HPV16)-transformed syngeneic cells, mice were immunized with the HPV16-E7 peptide in incomplete Freund's adjuvant with or without an anti-CD40 monoclonal, or left untreated. (Data from Diehl L. *et al.* (1999) *Nature Medicine* 5, 774, reproduced with permission.)

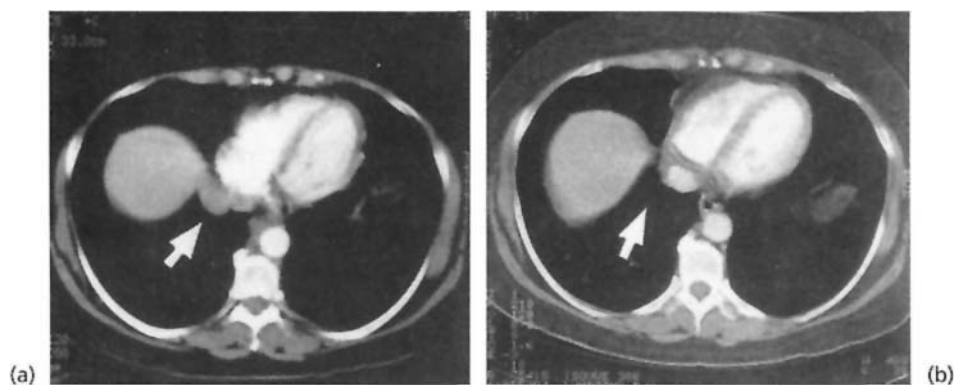


Figure 18.11. Clinical response to autologous vaccine utilizing dendritic cells pulsed with idiotype from a B-cell lymphoma. Computed tomography scan through patient's chest (a) prevaccine and (b) 10 months after completion of three vaccine treatments. The arrow in (a) points to a paracardiac mass. All sites of disease had

resolved and the patient remained in remission 24 months after beginning treatment. (Photography kindly supplied by Professor R. Levy from the article by Hsu F.J. *et al.* (1996) *Nature Medicine* 2, 52; reproduced by kind permission of Nature America Inc.)

100-fold and transfection with RNA and recombinant vectors such as fowlpox. Second, the procedure is cumbersome and costly but, if it becomes common, it will be streamlined and, anyway, the costs must be set against the expenses of conventional therapy and the immeasurable benefit to the patient. Third, why does the administration of small numbers of antigen-pulsed DCs induce specific T-cell responses and tumor regression in patients in whom both the antigen and DCs are already plentiful? The suggestion has been made that DCs in or near malignant tissues may be defective, perhaps due to vascular endothelium growth factor (VEGF) or IL-10 secretion by the tumor. It will certainly be interesting to see whether direct treatment of patients with Flt3-ligand plus accessory cytokines or intratumoral transfection with MIP-3 α (CCL20) (cf. p. 187), which attracts immature DCs, can initiate an antitumor response through maturation and activation of endogenous DCs.

Treatment of leukemia

Radical cytoablative treatment of leukemia patients using radiochemotherapy will destroy bone marrow stem cells. These can be removed prior to treatment, purged of any leukemic cells with cytotoxic antibodies, and reinfused subsequently to 'rescue' the patient (figure 18.12). However, not all leukemic cells are eliminated by this treatment, and a more effective strategy is transplantation of *allogeneic* bone marrow from reasonably MHC-compatible donors which exerts an important, albeit not completely understood, graft-vs-leukemia effect. Purging the bone marrow of T-cells to prevent graft-vs-host (g.v.h.) disease, which is a seri-

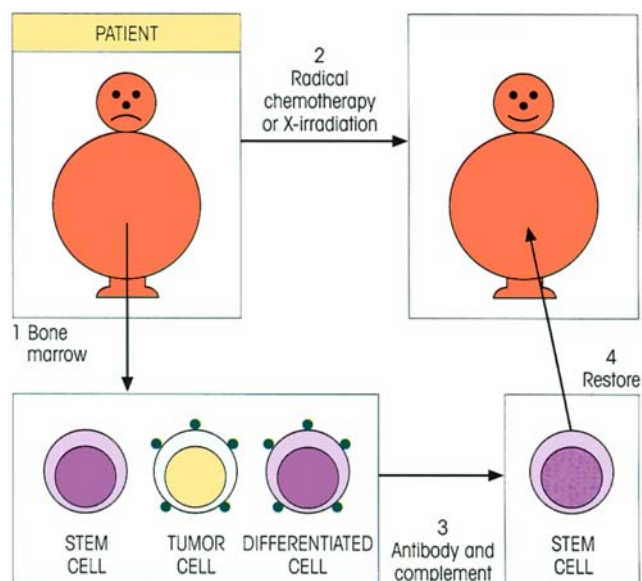
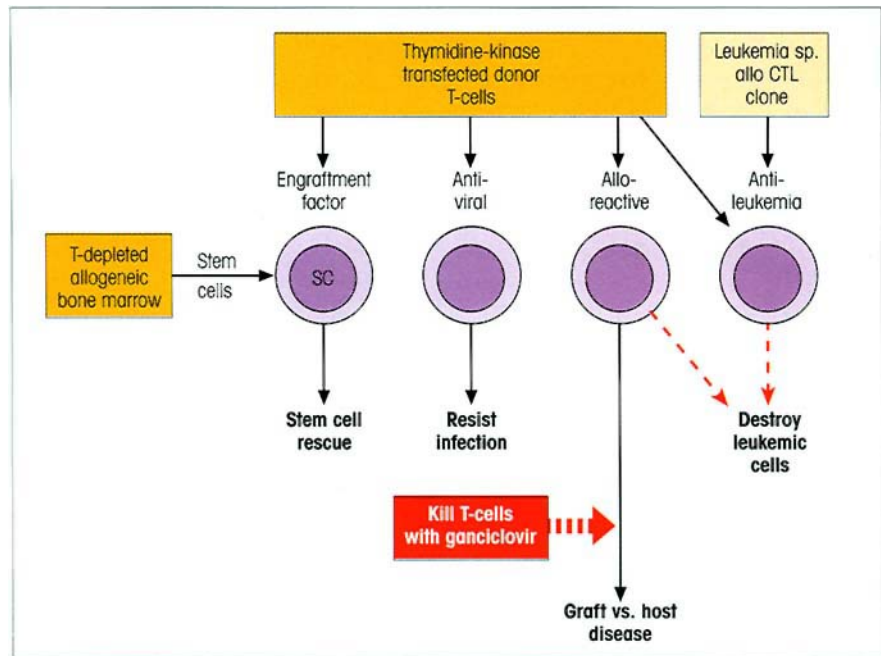


Figure 18.12. Treatment of leukemias by autologous bone marrow rescue. By using cytotoxic antibodies to a differentiation antigen (●) present on leukemic cells and even on other normal differentiated cells, but absent from stem cells, it is possible to obtain a tumor-free population of the latter which can be used to restore hematopoietic function in patients subsequently treated radically to destroy the leukemic cells. Another angle is positive selection of stem cells utilizing the CD34 marker.

ous complication of such transplants, would at the same time remove the prized antileukemic activity. A dilemma. 'Suicide gene therapy' could provide a solution as illustrated in figure 18.13. Stem cells from the T-cell-purged allogeneic bone marrow are given together with donor T-cells transfected with herpes simplex virus thymidine kinase. The T-cells provide

Figure 18.13. Treatment of leukemia with allogeneic bone marrow transfer. T-depleted allogeneic marrow provides the stem cells to 'rescue' the patient treated with cytoablative therapy. T-cells from the donor, transfected with thymidine kinase (TK), help engraftment, provide protection against infection and eliminate residual tumor cells by a graft-vs-leukemia effect. The alloreactive cells eventually produce graft-vs-host disease and can be eliminated by administration of ganciclovir. This is converted by the TK into a nucleoside analog which ultimately becomes toxic for dividing cells. The alternative shown is to destroy leukemic cells by supplementing purged allogeneic marrow with a leukemia peptide-specific CTL clone produced in third party T-cells. (Based on articles by Cohen J.L., Boyer O. & Klatzmann D. (1999) *Immunology Today* 20, 172; and Stauss H.J. (1999) *Immunology Today* 20, 180.)



factors which facilitate engraftment, defense against viral infection and the graft-vs-leukemia action at a time when the recipient patient will have a low tumor burden. With time, as graft-vs-host disease develops, the dividing aggressor donor T-cells can be switched off by administration of ganciclovir through the mechanism explained in the legend to figure 18.13.

An alternative which avoids g.v.h. disease altogether is to inject the purged bone marrow together with an allogeneic cytotoxic T-cell clone specific for a leukemia-associated peptide presented by the MHC allele of the prospective recipient patient. This usually works because the residues on the MHC helices which contact the T-cell receptor are relatively conserved (unlike those within the groove), so that the allo-T-cells can recognize the MHC-peptide complex from the leukemia. Potential targets are cyclin-D1 and mdm-2 which are overexpressed in tumor cells and, in leukemic cells in particular, the transcription factors WT-1 and GATA1 and the differentiation antigens myeloperoxidase and CD68, which are expressed exclusively in hematopoietic cells and are likely to have established tolerance in the patient but not in the allogeneic CTL donor (who will have been exposed to different processed peptides) (cf. p. 353). A rather masterful development would be to transfect the recipient with the genes encoding the T-cell receptor of the allo-CTL clone. Again, looking to the future, transfection of T-cells *in vitro* with a humanized scFv-Fcγ-transmembrane segment-CD3ζ construct

has been shown to render them cytotoxic for cells expressing the surface antigen.

Therapy with monoclonal antibodies

The strategies

Antibodies reacting with antigens on the surface of tumor cells can protect the host by complement-mediated opsonization and lysis (modified by host complement regulatory proteins) and through recruitment of macrophage and NK ADCC function by engagement of FcγRIII receptors, although for macrophages this is partially countered by inhibitory FcγRII signals. These FcR-bearing cells serve not only as cytotoxic effectors but also as multivalent surfaces which hyper-cross-link antibody-coated target cells so providing, in many cases, a transmembrane signal which leads to apoptosis or premature exit from the cell cycle. This effect appears to sensitize neoplastic cells to irradiation and DNA-damaging chemotherapy and holds out the exciting prospect of novel synergistic treatments whose efficacy may be enhanced by the increased immunogenicity of the dying cells.

Immunologists have long been bemused by the idea of eliminating tumor cells by specific antibody linked to a killer molecule and there is a truly impressive array of ingenious initiatives. It is axiomatic that multimeric fragments bind much more avidly than monomeric fragments due principally to the lower

off-rates (cf. p. 87), and that constructs in the 60–120 kDa range are optimal for targeting solid tumors—too large and penetration is difficult, too small and kidney secretion is excessively fast. Monovalent fragments include Fv, scFv selected by antigen from phage libraries (cf. p. 124) and V_H domains based on the large CDR loops of the camels and llamas. For polymers we have bivalent and bispecific (think about the difference) diabodies (cf. p. 125), trivalent and trispecific triabodies, even tetrabodies, and Fabs have been linked into dimers or trimers.

Radioimmunotherapy, using isotopes such as ^{90}Y , ^{111}In or ^{131}I linked to antibody, delivers doses of radiation to tumor tissue which would be impossibly toxic with external beam sources. Results are even better when used in synergy with chemotherapy. Immunitoxins represent another class of magic bullet in which the toxin of plant or bacterial origin is attached to an antibody or growth factor; after binding to the cell surface, the toxin is internalized and kills the cell by catalytic inhibition of protein synthesis. Along the same lines, internalizing a photosensitizer renders the cells vulnerable to photodynamic therapy. Delivery of anticancer drugs by attachment to antibodies would seem to be an obvious goal, but a more sophisticated strategy involves the grand sounding antibody-directed enzyme-prodrug therapy (ADEPT). After injection of the prodrug, each antibody–enzyme conjugate attached to its cell target produces a large number of small toxic molecules which can diffuse into the tumor mass and also give a bystander effect on adjacent tumor cells even if they express only low levels of antigen. Aminopeptidase which acts on 2-L-pyroglutamyl methotrexate to liberate methotrexate is one of the conjugates used.

Clinical results

Useful responses to unmodified monoclonal antibodies have been chalked up. These include anti-Her-2 (an epidermal growth factor receptor also known as erbB2 and neu) for metastatic breast cancer, anti-CD52 (Campath-1H) for a range of hematological tumors and 17-1A for colorectal cancer.

^{131}I -labeled anti-CD20 kills B-cell lymphomas by ADCC and by signaling-induced apoptosis, acting in synergy with DNA damage due to γ -radiation, with perhaps a contribution from enhanced tumor immunogenicity. Therapy with this agent has been spectacularly successful; in one long-term trial, injection of the radiolabeled anti-CD20 into lymphoma patients prepared by myeloablation with stem cell rescue gave major responses in 86% with remissions lasting from

27 to 87 months. Likewise, radionuclide conjugates of a humanized anti-CD33 score very highly in myeloid leukemia and can root out large tumor burdens. Trials of ^{90}Y -anti-MUC-1 in ovarian cancer provide further encouragement for the intrepid immunotherapists.

Attack on the tumor blood supply

For solid tumors, the focus is upon two main targets. The first would be **minimal residual micrometastases in the bone marrow** which occur in one-third to one-half of patients with epithelial cancer after curative radical treatment of the primary lesion. The second would be the **reactive tissue evoked by the malignant process**, such as stromal fibroblasts expressing the F19 glycoprotein and newly formed blood vessels.

Tumors generally cannot grow beyond the size of 1 mm in diameter without the support of blood vessels and, because the tumor is new tissue, its blood vessels will also have to be new. These vessels form through the process known as **angiogenesis**, the sprouting of new blood vessels from existing ones, but they are biochemically and structurally different from normal resting blood vessels and so provide differential targets for therapeutic monoclonal antibodies, even though the cancer cells themselves in a solid tumor are less vulnerable to antibodies directed to specific antigens on their surface. Thus, receptors for VEGF and Eph, oncofetal fibronectin, matrix metalloproteases MMP-2 and MMP-9 and the pericyte markers aminopeptidase A and the NG2 proteoglycan are all highly and selectively expressed in vasculature undergoing angiogenesis.

A noteworthy maneuver, which is unexpectedly successful, is to identify peptides which home specifically to the endothelial cells of certain tumors by injecting peptide phage libraries *in vivo*. One of the panel of peptide motifs which has emerged from this probing strategy includes RGD in the cyclic peptide CDCRGD-CFC, a selective binder of the $\alpha_v\beta_3$ - and $\alpha_v\beta_5$ -integrins known to be upregulated in angiogenic tumor endothelial cells. For therapeutic exploitation, these peptides can be linked to appropriate drugs, such as doxorubicin, or a pro-apoptotic peptide. Overall, there are undoubtedly a substantial number of targets for the ‘magic bullets’.

IMMUNODIAGNOSIS OF SOLID TUMORS

Circulating and cellular tumor markers

Analysis of blood for the oncofetal antigens, α -fetoprotein in hepatoma and carcinoembryonic anti-

gen in tumors of the colon, has provided valuable diagnostic information, but enthusiasm has been slightly curtailed by the knowledge that there is a high incidence of so-called 'false positives'. Reappearance of these proteins after surgical removal of the primary is strongly indicative of fresh tumor growth. A hefty increase in the ratio of free to bound prostate-specific antigen (PSA) in the blood may signal cancer of the prostate. The GM1 monosialoganglioside has been demonstrated in the blood of 96% of patients with pancreatic carcinoma and 64% of patients with colorectal carcinomas, as against 2% in normal subjects.

Identification of the cell type by monoclonal antibodies is of value for the diagnosis and treatment of an increasing number of tumors, including the lymphoproliferative disorders as discussed earlier (see p. 382).

Tumor imaging *in vivo*

The same principles which govern the localization of monoclonal antibodies for tumor therapy apply equally to imaging. Maximizing the binding to tumor relative to normal tissue and surrounding fluids is the name of the game. For example, the use of a bifunctional antibody which targets the tumor and an isotope chelator can be followed 24–120 hours later with the chelate-containing radionuclide which allows clearance of uncombined antibody.

The Thomson–Friedenreich (T) antigen (Gal β 1-3-GalNAc α -O-Ser), expressed in the mucins of various types of epithelial cancer, has proved to be a highly successful target for antibody imaging. So has the F19 glycoprotein associated with proliferating fibroblasts in the stroma of many carcinomas, as presumably the many antigens associated with tumor angiogenesis will prove to be.

Table 18.3. Detection of bone marrow micrometastases by staining for epithelial cytokeratin in colorectal cancer patients. (From data of Schlimok G. *et al.* (1990) *Journal of Clinical Oncology* 8, 831.)

Dukes' Stage		Positive reaction with mAb CK2 for cytokeratin in bone marrow aspirate	
		No. patients	% Positive
A	Limited to mucosa	3	0
B	Extending into muscularis propria	58	14
C	Involving local nodes	62	34
D	With distant metastatic spread	33	39

Detection of micrometastases in bone marrow

Because of the difficulty in detecting individual tumor cells in distant organs, the diagnosis of early disseminated cancer has not been possible, and attempts to identify earlier stages and to monitor the immunotherapy of early disease have been hindered. A major advance was made when micrometastases were demonstrated by immunocytochemistry in the bone marrow of patients with colorectal cancer and were related to more widespread disease (table 18.3) and a high relapse rate. The method involves scanning pelvic crest bone marrow aspirates taken at surgery for epithelial cells by staining for cytokeratin (cf. figure 18.7f) and proliferation markers such as the Ki67 nuclear antigen and receptors for transferrin and epidermal growth factor. Detection of micrometastases in the marrow of patients with small cell lung carcinoma also predicted early relapse.

SUMMARY

Changes on the surface of tumor cells

- Processed peptides derived from oncogenic viruses are powerful MHC-associated transplantation antigens.
- Some tumors express genes which are silent in normal tissues: sometimes they have been expressed previously in embryonic life (oncofetal antigens).
- Many tumors express weak antigens associated with point mutations in oncogenes such as *ras* and *p53*. Peptides

presented by heat-shock proteins 70 and 90 represent the unique chemically induced tumor antigens. The surface Ig on chronic lymphocytic leukemia (CLL) cells is a unique tumor-specific antigen.

- Dysregulation of tumor cells frequently causes structural abnormalities in surface carbohydrate structures.
- Some surface changes are a consequence of cell division *per se*.

(continued p. 394)

- The v6 and v10 exons of CD44 are intimately involved with metastatic potential. Loss of blood group A determinants leads to a poor prognosis.

Immune response to tumors

- T-cells generally mount effective surveillance against tumors associated with oncogenic viruses or UV induction which are strongly immunogenic.
- More weakly immunogenic tumors are not controlled by T-cell surveillance, although sometimes low-grade responses are evoked.
- NK cells probably play a role in containing tumor growth and metastases. They can attack MHC class I-negative tumor cells because the class I molecule imparts a negative inactivation signal to NK cells. The A-NK subset, which expresses high levels of adhesion molecules, is more cytolytic for fresh tumor cells.
- Tumors utilize a variety of mechanisms to escape host immune responses.

Unregulated development gives rise to lymphoproliferative disorders

- Deregulation of the *c-myc* protooncogene is a characteristic feature of many B-cell tumors.
- Chromosome translocations are common.
- Lymphoid malignancies show maturation arrest at characteristic stages in differentiation.
- The surface markers of leukemias and lymphomas identified by monoclonal antibodies are important aids in diagnosis. Most non-Hodgkin's lymphomas are of B-cell origin, are associated with EBV and express a monoclonal surface Ig.
- Multiple myeloma represents a malignant proliferation of a single clone of plasma cells producing a single 'M' band on electrophoresis. 10–20% have widespread amyloid deposits containing the variable region of the myeloma light chain.
- Waldenström's macroglobulinemia is produced by unregulated proliferation of a clone producing monoclonal IgM causing a marked rise in serum viscosity.
- Malignant lymphoid cells produce secondary immunodeficiency by suppressing differentiation of the corresponding normal lineage.

Approaches to cancer immunotherapy

- Immunotherapy is only likely to work after a tumor mass has been debulked.
- Innate immune mechanisms can be harnessed. Systemic IL-12 may be effective against minimal residual disease. IFN γ and β are very effective in the T-cell disorders, hairy cell leukemia and mycosis fungoides, less so but still

significant in Kaposi's sarcoma and various lymphomas; they may be used in synergy with other therapies. GM-CSF enhances proliferation and decreases leukemogenicity of murine myeloid leukemias.

- Cancer vaccines based on oncogenic viral proteins can be expected.
- Weakly immunogenic tumors provoke effective anti-cancer responses if given with an adjuvant, such as BCG, or if transfected with costimulatory molecules, such as B7 and cytokines IFN γ , IL-2, -4 and -7. CD8 CTLs are favored for the attack on solid tumors.
- A variety of potential protein targets have been identified and intense effort is being expended in the investigation of peptides as subunit vaccines. Their immunogenicity can be enhanced by complexing with heat-shock proteins and by accessory factors such as GM-CSF, CTLA-4 blockade and anti-CD40 stimulation.
- Naked DNA and self-replicating RNA vaccines are strong candidates.
- Powerful immunogens have been created by pulsing dendritic antigen-presenting cells with peptides from melanoma antigens and framework regions of CLL Ig.
- A graft-vs-leukemia effect is achieved by allogeneic CTLs or by allogeneic bone marrow transplantation with measures to limit graft-vs-host disease.
- Monoclonal antibodies conjugated to toxins or radionuclides can target tumor cells or antigens on new blood vessels or the reactive stromal fibroblasts associated with malignancy. Encouraging, even impressive, therapeutic results have been obtained with antibodies to CD20 in B-cell lymphoma, CD33 in myeloid leukemia, anti-MUC-1 in ovarian cancer and c-erbB2 overexpressed on breast cancers. Bifunctional antibodies can bring effectors such as NK and Tc close to the tumor target.

Immunodiagnosis of solid tumors

- Many circulating tumor markers are diagnostic, e.g. α -fetoprotein in hepatic carcinoma and carcinoembryonic antigen in colorectal carcinoma.
- Monoclonal antibody to tumor surfaces can provide a basis for imaging. Certain tumor mucins, the F19 glycoprotein on reactive stromal fibroblasts and VEGF on new blood vessels around the tumor are good targets.
- Detection of micrometastases in bone marrow by immunocytochemistry provides valuable information on prognosis and the efficacy of new therapies.

See the accompanying website (www.roitf.com) for multiple choice questions.

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Autoimmune diseases

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INTRODUCTION

The monumental repertoire of the adaptive immune system has evolved to allow it to recognize and ensnare virtually any shaped microbial molecules, either at present in existence or yet to come, and in so doing, has been unable to avoid the generation of lymphocytes which react with the body's own constituents. This is abundantly so for T-cells which actually depend upon a degree of self-recognition for positive selection to operate during their development within the thymus gland (p. 228). We have already discussed the mechanisms which exist to prevent these self-components from provoking an adaptive immune response but, as with all machinery, there is always a chance that these systems might break down, and the older the individual, the greater the chance of this happening. Notwithstanding the IgM low affinity **autoantibodies** (i.e. antibodies capable of reacting with 'self'-components) produced by CD5⁺ B-cells as part of the 'natural' antibody spectrum which we will discuss later, we are here concerned more with

autoimmune phenomena which appear in relation to certain defined human diseases. Ideally, we wish to apply the term '**autoimmune disease**' to those cases where it can be shown that the **autoimmune process contributes to the pathogenesis of the disease** rather than situations where apparently harmless autoantibodies are formed following tissue damage, e.g. heart antibodies appearing after a myocardial infarction. Yet the role of autoimmunity in many disorders is still not clearly defined, and it is as a matter of convenience that we will refer to all maladies firmly associated with autoantibody formation as '**autoimmune diseases**', except where it can be shown that the immunological phenomena are purely secondary findings.

THE SCOPE OF AUTOIMMUNE DISEASES

The spectrum of autoimmune diseases

These disorders may be looked upon as forming a spectrum. At one end we have '**organ-specific diseases**'

with organ-specific autoantibodies. **Hashimoto's disease** of the thyroid is an example: there is a specific lesion in the thyroid involving infiltration by mononuclear cells (lymphocytes, macrophages and plasma

cells), destruction of follicular cells and germinal center formation, accompanied by the production of circulating antibodies with absolute specificity for certain thyroid constituents (Milestone 19.1).

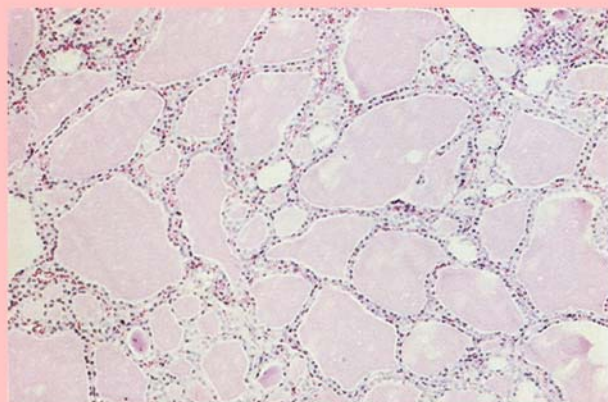
Milestone 19.1 — The Discovery of Thyroid Autoimmunity

Although Dacie's studies on red cell autoantibodies in certain forms of hemolytic anemia were amongst the earliest to implicate autoimmunity in the pathogenesis of disease, a direct link to disorders affecting whole organs was not established until 1956 when three major papers on thyroid autoimmunity appeared.

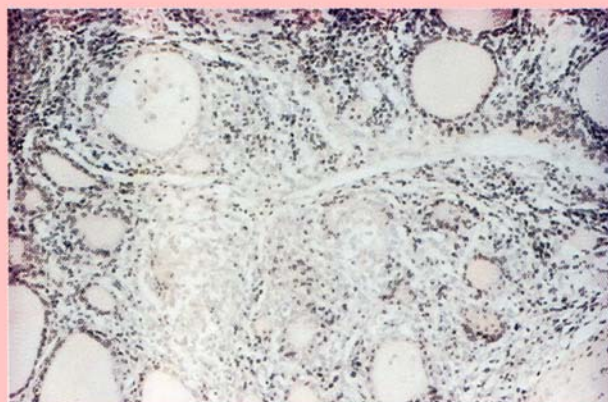
In an attempt to confirm Paul Ehrlich's concept of 'horror autotoxicus'—the body's dread of making antibodies to self—Rose and Witebsky immunized rabbits with rabbit thyroid extract in complete Freund's adjuvant. To what one might hazard was Witebsky's dismay and Rose's delight, this procedure resulted in the production of

thyroid autoantibodies and chronic inflammatory destruction of the thyroid gland architecture (figure M19.1.1a and b).

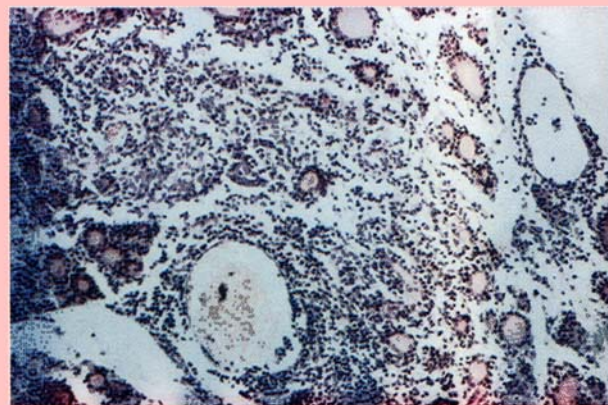
Having noted the fall in serum γ -globulin which followed removal of the goiter in Hashimoto's thyroiditis and the similarity of the histology (figure M19.1.1c) to that of Rose and Witebsky's rabbits, Roitt, Doniach and Campbell tested the hypothesis that the plasma cells in the gland might be making an autoantibody to a thyroid component, so causing the tissue damage and chronic inflammatory response. Sure enough, the sera of the first patients tested had precipitating antibodies to an autoantigen in



(a)



(b)



(c)

Figure M19.1.1. Experimental autoallergic thyroiditis. (a) The follicular architecture of the normal thyroid. (b) Thyroiditis produced by immunization with rat thyroid extract in complete Freund's adjuvant; the invading chronic inflammatory cells have destroyed the follicular structure. (Based on the experiments of Rose N.R. & Witebsky E. (1956) Studies on organ specificity. V. Changes in the thyroid gland of rabbits following active immunization with rabbit thyroid extracts. *Journal of Immunology* 76, 417.) (c) Similarity of lesions in spontaneous human autoimmune disease to those induced in the experimental model. Other features of Hashimoto's disease such as the eosinophilic metaplasia of acinar cells (Askenazy cells) and local lymphoid follicles are not seen in this experimental model, although the latter occur in the spontaneous thyroiditis of Obese strain chickens.

(continued p. 398)

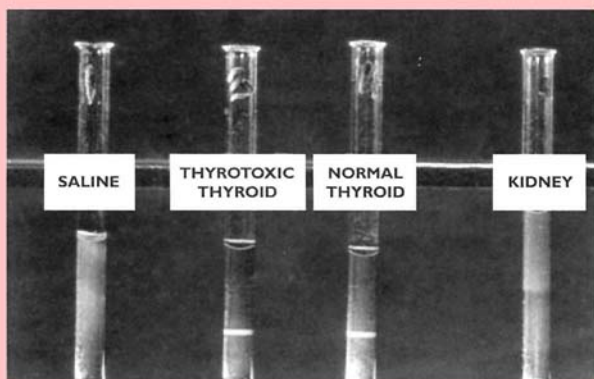


Figure M19.1.2. Thyroid autoantibodies in the serum of a patient with Hashimoto's disease demonstrated by precipitation in agar. Test serum is incorporated in agar in the bottom of the tube; the middle layer contains agar only, while the autoantigen is present in the top layer. As serum antibody and thyroid autoantigen diffuse towards each other, they form a zone of opaque precipitate in the middle layer. Saline and kidney extract controls are negative. (Based on Roitt I.M., Doniach D., Campbell P.N. & Hudson R.V. (1956) Autoantibodies in Hashimoto's disease. *Lancet* ii, 820.)

normal thyroid extracts which was soon identified as thyroglobulin (figure M19.1.2).

In far off New Zealand (depending on your geographical location!), Adams and Purves, in seeking a circulating factor which might be responsible for the hyperthyroidism of Graves' thyrotoxicosis, injected patient's serum into guinea-pigs whose thyroids had been prelabeled with ^{131}I , and followed the release of radiolabeled material from the gland with time. Whereas the natural pituitary

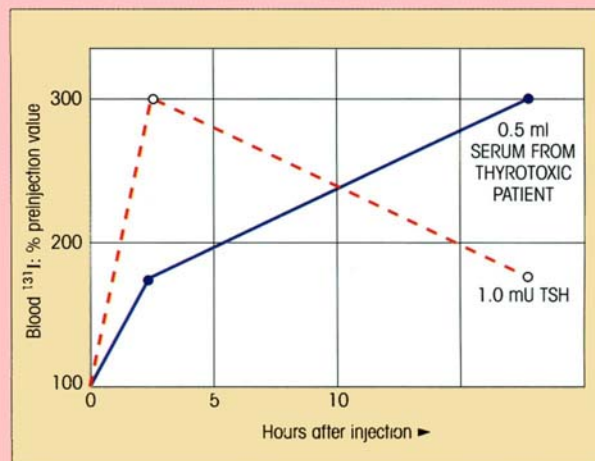


Figure M19.1.3. The long-acting thyroid stimulator in Graves' disease. Injection of TSH causes a rapid release of ^{131}I from the prelabeled animal thyroid in contrast to the prolonged release which follows injection of serum from a thyrotoxic patient. (Based on Adams D.D. & Purves H.D. (1956) Abnormal responses in the assay of thyrotrophin. *Proceedings of the University of Otago Medical School* 34, 11.)

thyroid-stimulating hormone (TSH) produced a peak in serum radioactivity some 4 hours or so after injection of the test animal, serum from thyrotoxic patients had a prolonged stimulatory effect (figure M19.1.3). The so-called *long-acting thyroid stimulator* (LATS) was ultimately shown to be an IgG mimicking TSH through its reaction with the TSH receptor but differing in its time-course of action, largely due to its longer half-life in the circulation.

Moving towards the center of the spectrum are those disorders where the lesion tends to be localized to a single organ but the antibodies are nonorgan-specific. A typical example would be **primary biliary cirrhosis** where the small bile ductule is the main target of inflammatory cell infiltration but the serum antibodies present—mainly mitochondrial—are not liver-specific.

At the other end of the spectrum are the '**nonorgan-specific or systemic autoimmune diseases**' broadly belonging to the class of rheumatological disorders, exemplified by **systemic lupus erythematosus (SLE)**, where both lesions and autoantibodies are not confined to any one organ. Pathological changes are widespread and are primarily lesions of connective tissue with fibrinoid necrosis. They are seen in the skin (the 'lupus' butterfly rash on the face is characteristic),

kidney glomeruli, joints, serous membranes and blood vessels. In addition, the formed elements of the blood are often affected. A bizarre collection of autoantibodies is found, some of which react with the DNA and other nuclear constituents of all cells in the body.

An attempt to fit the major diseases considered to be associated with autoimmunity into this spectrum is shown in table 19.1.

Autoantibodies in human disease

At this stage in the discussion it may be of value to have a more precise account of the major autoantibodies detected in the different diseases to provide a framework for reference. Table 19.2 documents a list of these antibodies and the methods employed in their detection. The notes accompanying the table amplify specific

Table 19.1. Spectrum of autoimmune diseases.

ORGAN SPECIFIC	
	Hashimoto's thyroiditis Primary myxedema Graves' disease Pernicious anemia Autoimmune atrophic gastritis Addison's disease Premature menopause (few cases) Male infertility (few cases) Myasthenia gravis Insulin-dependent diabetes mellitus Goodpasture's syndrome Pemphigus vulgaris Pemphigoid Sympathetic ophthalmia Phacogenic uveitis Multiple sclerosis Autoimmune hemolytic anemia Idiopathic thrombocytopenic purpura Idiopathic leukopenia Primary biliary cirrhosis Active chronic hepatitis HBs—ve Ulcerative colitis Sjögren's syndrome Rheumatoid arthritis Scleroderma Wegener's granulomatosis Poly/dermatomyositis Discoid lupus erythematosus Systemic lupus erythematosus (SLE)
	↓
NONORGAN SPECIFIC	

points, while some of the tests are illustrated in figures 19.1, 6.9, 6.10 and 7.3. As antigens are characterized and become available in purified form, the convenient ELISA is becoming a dominant technique.

Overlap of autoimmune disorders

There is a tendency for more than one autoimmune disorder to occur in the same individual and when this happens the association is often between diseases within the same region of the autoimmune spectrum (cf. table 19.1). Thus patients with autoimmune thyroiditis (Hashimoto's disease or primary myxedema) have a much higher incidence of pernicious anemia than would be expected in a random population matched for age and sex (10% as against 0.2%). Conversely, both thyroiditis and Graves' disease are diagnosed in pernicious anemia patients with an unexpectedly high frequency. Other associations are seen between Addison's disease and autoimmune thyroid disease and occur in the rare cases of juveniles with pernicious anemia and polyendocrinopathy which includes Addison's disease, hypoparathyroidism, diabetes and thyroiditis.

There is an even greater overlap in serological findings. Thirty per cent of patients with autoimmune thyroid disease have concomitant parietal cell antibodies

in their serum. Conversely, thyroid antibodies have been demonstrated in up to 50% of pernicious anemia patients. It should be stressed that these are not cross-reacting antibodies. The thyroid-specific antibodies will not react with stomach and vice versa. When a serum reacts with both organs it means that two populations of antibodies are present, one with specificity for thyroid and the other for stomach.

At the nonorgan-specific end of the spectrum, systemic autoimmune disease such as SLE is clinically associated with rheumatoid arthritis and several other disorders which are themselves uncommon: hemolytic anemia, idiopathic leukopenia and thrombocytopenic purpura, dermatomyositis and Sjögren's syndrome. Antinuclear antibodies and antiglobulin (rheumatoid) factors are a general feature.

Sjögren's syndrome occupies an interesting position (table 19.3); aside from the clinical and serological features associated with systemic disease mentioned above, characteristics of an organ-specific disorder are evident. Antibodies reacting with salivary ducts are demonstrable and there is an abnormally high incidence of thyroid autoantibodies; histologically the affected lacrimal and salivary glands reveal changes of a similar nature to those seen in Hashimoto's disease, namely a replacement of the glandular elements by patchy lymphocytic and plasma cell granulomatous tissue. Associations between diseases at the two ends of the spectrum have been reported, but, as might be predicted from the serological data (table 19.3), they are not common.

Patients with organ-specific disorders are slightly more prone to develop cancer in the affected organ, whereas generalized lymphoreticular neoplasia shows up with uncommon frequency in nonorgan-specific disease.

Animal models of autoimmune disease

Both spontaneous and induced animal models have given tremendous insights into the nature of human autoimmune disease and, to assist our discussions, we felt it would be helpful to list them (table 19.4).

NATURE AND NURTURE

Genetic factors in autoimmune disease

Autoimmune phenomena tend to aggregate in certain families. For example, the first degree relatives (sibs, parents and children) of patients with Hashimoto's disease show a high incidence of thyroid autoantibodies (figure 19.2) and of overt and subclinical thyroiditis.

Table 19.2. Autoantibodies in human disease.

DISEASE	ANTIGEN	DETECTION OF IMMUNOLOGICAL REACTIVITY
Hashimoto's thyroiditis Primary myxedema	Thyroglobulin Thyroid peroxidase: Cytoplasmic Cell surface	Precipitins; passive hemagglutination; ELISA IFT on unfixed thyroid; passive hemagglutination; ELISA IFT on viable thyroid cells; C'-mediated cytotoxicity
Graves' disease	Cell surface TSH receptors	Bioassay—stimulation of mouse thyroid <i>in vivo</i> ; blocking combination TSH with receptors; stimulation adenyl cyclase
Pernicious anemia	'Growth' receptors Intrinsic factor	Induction of cell division in thyroid fragments Neutralization; blocking combination with vit-B ₁₂ ; binding to intrinsic factor-B ₁₂ by coprecipitation
Addison's disease	Parietal cell H ⁺ -K ⁺ ATPase	IFT on unfixed gastric mucosa
Premature onset of menopause ¹	Cytoplasm adrenal cells (17 α -/21-hydroxylase)	IFT on unfixed adrenal cortex
Male infertility (some) ²	Cytoplasm steroid-producing cells	IFT on adrenal and interstitial cells of ovary and testis
Insulin-dependent diabetes ³	Spermatozoa	Sperm agglutination in ejaculate
Type B insulin resistance with acanthosis nigricans	Cytoplasm of islet cells Insulin, GAD and ICA512 Insulin receptor	IFT on unfixed human pancreas ELISA Block hormone binding to receptor
Atopic allergy (some)	β -Adrenergic receptor	Blocking radioassay with hydroxybenzylpindolol
Myasthenia gravis	Skeletal and heart muscle Acetylcholine receptor	IFT on skeletal muscle Blocking or binding radioassay with α -bungarotoxin
Lambert-Eaton syndrome	Ca ²⁺ channels in nerve endings	IgG produces neuromuscular defects in mice
Multiple sclerosis ⁴	Brain incl. MBP	MBP-reactive T-cells
Goodpasture's syndrome	Glomerular and lung basement membrane	Linear staining by IFT of kidney biopsy with fluorescent anti-IgG Radioimmunoassay with purified Ag; ELISA
Pemphigus vulgaris	Desmosomes between prickle cells in epidermis (cadherin)	IFT on skin
Pemphigoid	Basement membrane	IFT on skin
Phacogenic uveitis	Lens	Passive hemagglutination
Sympathetic ophthalmia	Uvea	(Delayed skin reaction to uveal extract)
Autoimmune hemolytic anemia ⁵	Erythrocytes	Coombs' antiglobulin test
Idiopathic thrombocytopenic purpura	Platelets	Shortened platelet survival <i>in vivo</i>
Primary biliary cirrhosis	Mitochondria (pyruvate dehydrogenase)	IFT on mitochondria-rich cells (e.g. distal tubules of kidney)
Active chronic hepatitis (HBV & HCV -ve)	Smooth muscle/nuclear lamins/nuclei Kidney/liver microsomes (cyt P450)	IFT (e.g. on gastric mucosa)
Ulcerative colitis	Colon 'lipopolysaccharide'	IFT (kidney)
	Colon epithelial cell surface protein	IFT; passive hemagglutination (cytotoxic action of lymphocytes on colon cells) ADCC on colon cancer cell line Ab data in this disease not universally accepted
Sjögren's syndrome ⁶	SS-A(Ro) SS-B(La) Ducts/mitochondria/nuclei/thyroid	IFT; gel precipitation; ELISA IFT
Rheumatoid arthritis ⁷	IgG IgG	Antiglobulin (rheumatoid factor) tests Antiglobulin test; latex agglutination; sheep red cell agglutination test (SCAT; commercial product, RAHA test) and ELISA; agalacto-glycoform
Discoid lupus erythematosus	Collagen	Passive hemagglutination
Scleroderma ⁸	Nuclear/IgG Nuclear/IgG/centromere Nuclear/IgG/Scl-70	IFT/antiglobulin test IFT
Dermatomyositis ⁹	Nuclear/IgG/Jo-1	IFT; countercurrent electrophoresis; ELISA
Mixed connective tissue disease ¹⁰	Extractable nuclear	IFT; countercurrent electrophoresis; ELISA
Systemic lupus erythematosus	DNA ¹¹ snRNP (Sm & ribonucleoprotein) Nucleoprotein Array of other Ag including formed elements of blood/IgG	ELISA; IFT on Crithidia IFT; gel precipitation techniques; ELISA IFT
Wegener's granulomatosis	Cardiolipin/ β 2-glycoprotein 1 Neutrophil cytoplasm (ANCA; myeloperoxidase/serine proteinase) ¹²	Radioimmunoassay IFT on alcohol fixed polymorphs; ELISA

IFT, immunofluorescent test (cf. figure 7.3); ELISA, enzyme-linked immunosorbent assay (cf. figure 6.10).

Table 19.3. Organ-specific and nonorgan-specific serological interrelationships in human disease.

DISEASE	% POSITIVE REACTIONS FOR ANTIBODIES TO:			
	THYROID*	STOMACH*	NUCLEI*	IgG†
Hashimoto's thyroiditis	99.9	32	8	2
Pernicious anemia	55	89	11	
Sjögren's syndrome	45	14	56	75
Rheumatoid arthritis	11	16	50	75
SLE	2	2	99	35
Controls‡	0–15	0–16	0–19	2–5

*Immunofluorescence test.

†Rheumatoid factor classical tests.

‡Incidence increases with age and females > males.

Parallel studies have disclosed similar relationships in the families of pernicious anemia patients, in that gastric parietal cell antibodies are prevalent in the relatives who are wont to develop achlorhydria and atrophic gastritis. Turning to SLE, disturbances of immunoglobulin synthesis and a susceptibility to develop 'connective tissue diseases' have been reported, but there are some conflicting accounts still not resolved.

These familial relationships could be ascribed to environmental factors such as infective microorganisms, but there is powerful evidence that important genetic components must be involved. The data on

twins is unequivocal. When thyrotoxicosis or insulin-dependent diabetes mellitus (IDDM) occurs in twins, there is a far greater concordance rate (i.e. both twins affected) in identical than in nonidentical twins. Second, we have already noted that lines of animals have been bred which spontaneously develop autoimmune disease (table 19.4). In other words, **the autoimmunity is genetically programmed**. There is an Obese line of chickens with autoimmune thyroiditis, the Nonobese diabetic (NOD) mouse modeling human IDDM and the New Zealand Black (NZB) strain succumbing to autoimmune hemolytic anemia. The hybrid of NZB with another strain, the New Zealand White

Notes to table 19.2

1 Antibodies occur in the minority of patients with associated Addison's disease of the adrenal and are directed to the 17 α /21-hydroxylase, the cholesterol side-chain cleavage enzyme and a 51 kDa gonadal antigen.

2 Only a small percentage show agglutinins. Spermatozoa may be agglutinated head to head, tail to tail or joined through their midpiece. Seen also in a small percentage of infertile women.

3 Most if not all insulin-dependent diabetics have islet cell antibodies at some stage during the first year of onset but these tend to decline progressively. In contrast, islet cell antibodies in diabetic patients with an associated autoimmune polyendocrinopathy persist for many years. GAD (glutamic acid decarboxylase) Ab also occur in Stiff man syndrome. ICA512 is a protein tyrosine kinase.

4 MBP, myelin basic protein.

5 The Coombs' test involves the demonstration of bound antibody on the washed red cell by agglutination with an antiglobulin (cf. figure 16.16). Erythrocyte autoantibodies, which bind well over the temperature range 0–37°C ('warm' Ab), are mostly IgG; approximately 60% of cases are primary, the remainder being associated with other autoimmune disorders, e.g. SLE and ulcerative colitis. 'Cold' Ab, which react best over the range 0–20°C, are mostly IgM, and red cells coated with this Ab can often be agglutinated by anticomplement serum; approximately half are primary, the others being associated with *Mycoplasma pneumoniae* infection or generalized neoplastic disease of the lymphoreticular tissues.

6 Antibodies specifically reacting with the epithelium of salivary gland excretory ducts are demonstrable by immunofluorescence in over half the cases of secondary Sjögren's associated with RA or SLE. SS-A and SS-B antibodies give a speckled nuclear fluorescence pattern.

7 The main antiglobulin factors react with the Fc portion of IgG which is usually adsorbed on to latex particles (human IgG) or present in an antigen-antibody complex (sheep red cells coated with a subagglutinating dose of rabbit antibody). In the ELISA test, rabbit IgG is bound to a plastic tube, patient's serum added and the antiglobulin bound assessed by subsequent binding of labeled anti-human IgG or IgM (cf. figure 6.10). Rheumatoid factors specific for human IgG can be detected by this test using human Fc γ to coat the tubes and labeled anti-human Fd γ (the portion of the heavy chain in the Fab fragment) or IgM for the final stage.

8 In scleroderma (progressive systemic sclerosis) antinucleolar antibodies are frequently found. Scl-70 is topoisomerase 1.

9 Jo-1 is histidine tRNA synthetase.

10 This syndrome combines features of scleroderma, RA, SLE and dermatomyositis. The antigens are extractable from the nucleus and give a speckled fluorescence pattern.

11 Antibodies to single- or double-stranded DNA are assayed by a DNA-coated tube test.

12 Component of primary granule, probably serine proteinase III, gives cytoplasmic staining. In periarteritis nodosa, antibodies to myeloperoxidase give perinuclear staining in alcohol-fixed polymorphonuclear neutrophils (PMNs). Some sera react with azurocidin, a potent antibiotic. ANCA (antineutrophil cytoplasmic antibody) directed against bactericidal/permeability increasing protein is a marker for inflammatory bowel disease and primary sclerosing cholangitis.

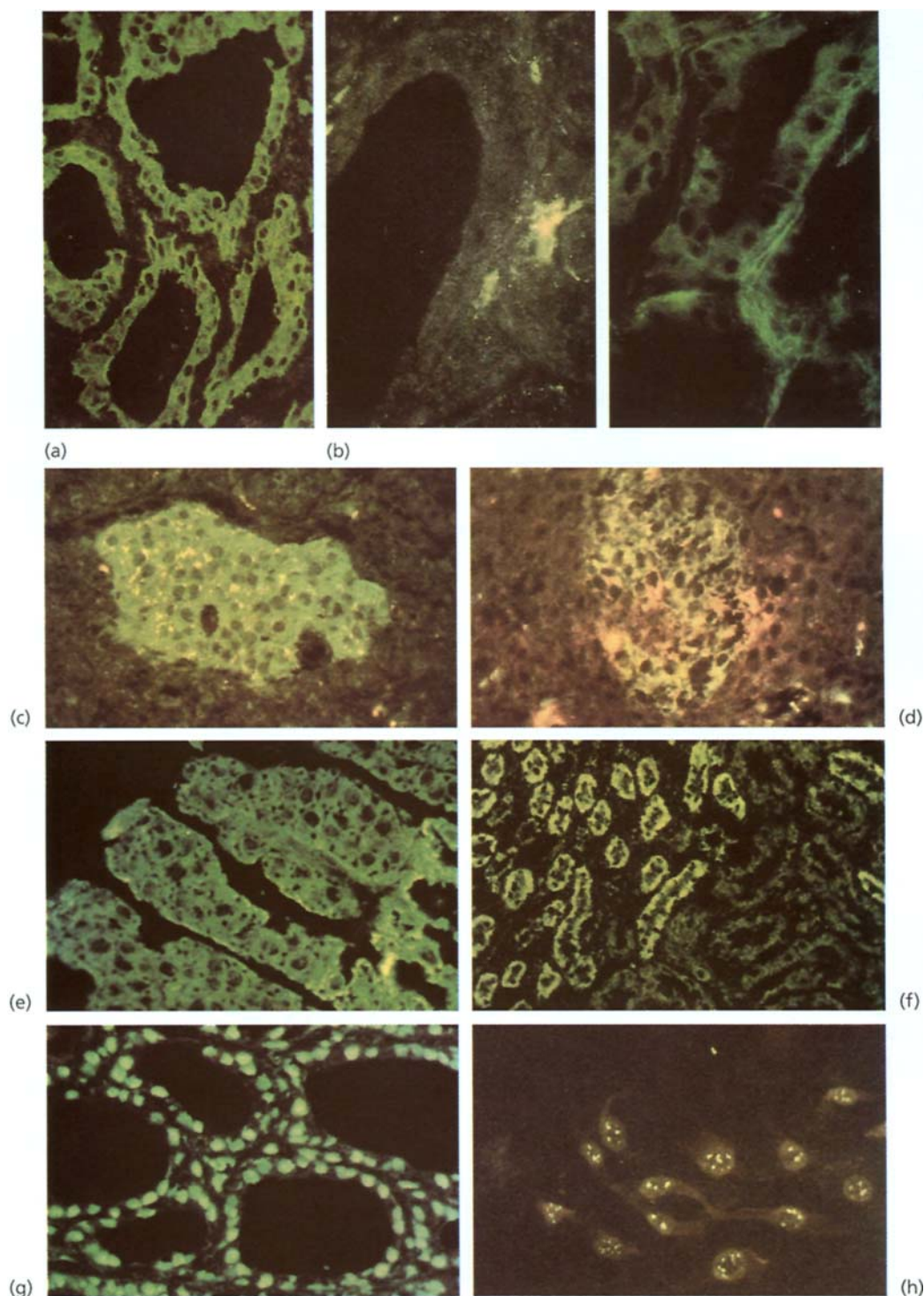


Figure 19.1. Fluorescent antibody studies in autoimmune diseases. (a) Thyroid peroxidase (thyroid microsomal) antibodies staining cytoplasm of acinar cells. (b) Human thyroid sections stained for MHC class II: *left*—normal thyroid with unstained follicular cells and an isolated strongly MHC class II-positive dendritic cell; *right*—Graves' disease (thyrotoxic) thyroid with abundant cytoplasmic MHC class II indicative of active synthesis. (c) Fluorescence of cells in the pancreatic islets of Langerhans' stained with serum from insulin-dependent diabetic. (d) The same, showing cells stained simultaneously for somatostatin (the yellow cells are stained with rhodamine

anti-somatostatin and fluorescein anti-human IgG which localizes the patient's bound autoantibody). (e) Serum of patient with Addison's disease staining cytoplasm of monkey adrenal granulosa cells. (f) Fluorescence of distal tubular cells of the kidney after reaction with mitochondrial autoantibodies. (g) Diffuse nuclear staining on a thyroid section obtained with nucleoprotein antibodies from a SLE patient. (h) Serum of a scleroderma patient staining the nucleoli of SV40-transformed human keratinocytes (K14) in monolayer culture. ((a), (c), (d), (e), (f) and (g) kindly provided by Prof. F. Bottazzo; (b) by Professor R. Pujol-Borrell; and (h) by Dr F.T. Wojnarowska.)

Table 19.4. Spontaneous and induced animal models of autoimmune disease.

		MODEL	AUTOIMMUNE DISEASE
ORGAN-SPECIFIC	SPONTANEOUS	Nonobese diabetic (NOD) mouse; BB rat NOD.H2 nd congenic Obese strain (OS) chicken; Buffalo rat	Insulin-dependent diabetes *Thyroiditis Thyroiditis
	INDUCED	**Complete Freund's adjuvant (CFA) incorporating brain CFA incorporating thyroid, adrenal, sperm, type II collagen, ACh-R, retinal S1 protein or g.b.m. Mouse bearing IFN γ -insulin promoter transgene Cross reaction: heterologous with autologous r.b.c. Coxsackie B with myosin Thymectomy in 2–4 day old mice Neonatal thymectomy + X-irradiation in rats HgCl ₂ in rats	Autoimmune encephalomyelitis Destruction of cell/tissue bearing relevant antigen Diabetes Anemia Myocarditis Widespread organ-specific Thyroiditis 'Goodpasture's'
SYSTEMIC	SPONTANEOUS	New Zealand Black (NZB) mouse strain NZB x W, BXSB p21 Knockout mouse MRL/lpr Mothaten mouse strain	Autoimmune hemolytic anemia SLE SLE SLE, arthritis Widespread fatal systemic disease
	INDUCED	Parent bone marrow into F1 mice CFA incorporating anti-DNA idiotype CFA incorporating TB hsp	G.v.h., pseudo SLE SLE Adjuvant arthritis

*Fed high iodine diet.

**Antigen emulsified in water/oil mixture containing killed tubercle bacilli or derivative.

ACh-R, acetyl choline receptor; r.b.c., erythrocytes; g.b.m., glomerular basement membrane.

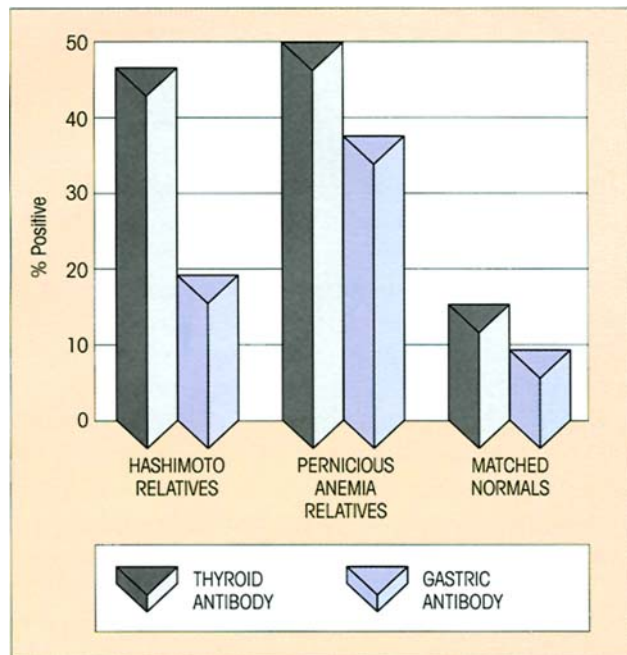


Figure 19.2. The high incidence of thyroid and gastric autoantibodies in the first degree relatives of patients with Hashimoto's disease or pernicious anemia. Note the overlap of gastric and thyroid autoimmunity and the higher incidence of gastric autoantibodies in pernicious anemia relatives. In general, titers were much higher in patients than in controls. (Data from Doniach D. & Roitt I.M. (1964) *Seminars in Haematology* 1, 313.)

(B×W hybrid), actually develops antinuclear antibodies including anti-dsDNA and a fatal immune complex-induced glomerulonephritis, key features of human SLE.

These diseases are **genetically complex**. Genome-wide searches for mapping the genetic intervals containing genes for predisposition to disease by linkage to the many thousand microsatellite markers (polymorphic variable numbers of tandem repeats, VNTR) have so far identified 20 such regions for IDDM in NOD mice and some 25 for murine SLE.

Generally speaking, a genetic predisposition to sustained inflammatory responses and loss of tolerance to self are major contributory factors. Dominant amongst the genetic associations with autoimmune diseases is **linkage to the major histocompatibility complex (MHC)**; of the many examples, we may recall the increased risk of IDDM for DQ8 individuals, and the higher incidence of DR3 in Addison's disease and of DR4 in rheumatoid arthritis (table 17.1, p. 366). Figure 19.3 shows a multiplex family with IDDM in which the disease is closely linked to a particular HLA haplotype. Another pointer to the central role of class II structure in determining T-cell responsiveness to self derives from the inability of the NOD mouse to develop pancreatic autoimmunity when just a single amino acid residue in the α -helix of the H-2 β chain is altered by

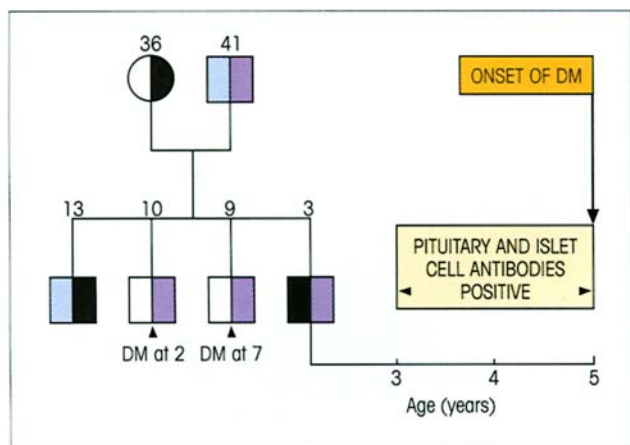


Figure 19.3. HLA haplotype linkage and onset of insulin-dependent diabetes (DM). Haplotypes: □ A3, B14, DR6; ■ A3, B7, DR4; □ A28, B51, DR4; and ■ A2, B62, C3, DR4. Disease is linked to possession of the A2, B62, C3, DR4 haplotype. The 3-year-old brother had complement-fixing antibodies to the islet cell surface for 2 years before developing frank diabetes indicative of the lengthy pathological process preceding disease. (Data provided by Prof. G.F. Bottazzo.)

introduction of a transgene. The close relationship to MHC is not altogether unexpected given that, as we shall see, autoimmune diseases are T-cell dependent and most T-cell responses are MHC restricted.

Amongst the plethora of non-MHC-linked loci may be genes responsible for the editorial control of rearranged Ig variable region genes encoding high avidity anti-DNA. Others may control the pattern of cytokine secretion affecting the milieu in early SLE which permits polyclonal B-cell activation, or influence the balance of Th1/Th2 subsets which could enhance susceptibility to IDDM or lead to resistance in otherwise predisposed subjects. Mice lacking the *p21* gene, which is a cell cycle regulator in the immune system, develop antibodies to dsDNA and other features of SLE. The gene maps within a recently defined MHC susceptibility locus with strong evidence for disease linkage. A mutation in the *IL-2* gene not affecting its functional ability to produce proliferation may be a candidate for *Idd-3* contributing to spontaneous diabetes in the NOD mouse and for *Aod-2* controlling autoimmune ovarian dysgenesis provoked by neonatal thymectomy. Polymorphism at a candidate locus identified in more than one autoimmune disease is interesting, a good example being *CTLA-4*, recently linked to IDDM and Graves' disease. *CTLA-4* mediates antigen-specific apoptosis capable of clonally deleting previously activated T-cells, and other 'apoptotic genes' such as *Fas*, *FasL* and *Bcl-2* are all involved in different autoimmune disorders. With so many cells dying

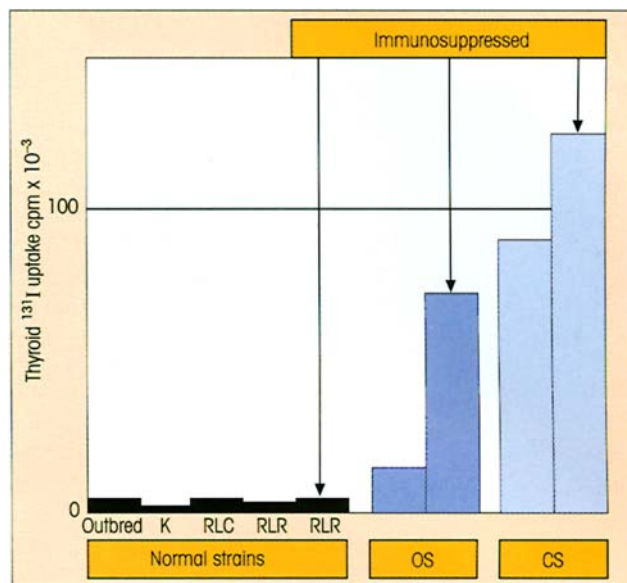


Figure 19.4. Abnormality in the Obese strain (OS) chicken thyroid. ^{131}I uptake in (OS) chickens and the related Cornell strain (CS) from which they were derived, compared with normal strains. Endogenous TSH production was suppressed by administration of thyroxine so that one was measuring TSH-independent ^{131}I uptake. Immune suppression showed that the abnormally high uptake was not due to immunological stimulation. (From Sundick R.S., Bagchi N., Livezey M.D., Brown T.R. & Mack R.E. (1979) Abnormal thyroid regulation in chickens with autoimmune thyroiditis. *Endocrinology* (Baltimore) 105, 493.)

naturally from apoptosis, it is clearly desirable for them to be immunologically inert, a state that may be achieved by the activation of transglutaminase which causes protein cross-linking and increased phagocytic removal.

Appropriate breeding experiments disclose that the genes predisposing to aggressive autoimmunity, on the one hand, are distinct from those which determine which autoantigens are involved, on the other. The 'autoimmunity genes' contribute the common element underlying the overlaps in autoantibodies and disease discussed above, although within this group the genes which predispose to organ-specific disease must be different from those in nonorgan-specific disorders (as judged by the minimal overlap between them).

Evidence for 'autoantigen and tissue selection' genes derives not only from breeding experiments with NZB×W mice showing separate control of red cell and nuclear antibodies, but also from genetic analysis of Obese chickens which has delineated an influence of the MHC, abnormalities in regulatory T-cell control and a defect in the thyroid gland expressed as an abnormally high uptake of ^{131}I (figure 19.4). Unlike normal thyroid cells, Hashimoto thyrocytes in culture

display Fas molecules on their surface and, since FasL is constitutively expressed, this leads of course to mutual apoptotic homicidal death. However, since destruction of the gland during the course of disease is a prolonged process, it seems likely that, *in vivo*, this catastrophic scenario may be tempered by anti-apoptotic factors such as bcl-2. In the family studies described above (figure 19.2), there must be additional genes which are organ-related, in that relatives of patients with pernicious anemia are more prone to gastric autoimmunity than are members of Hashimoto kindreds. Further to this point, analysis has mapped the *IDDM-2* diabetes susceptibility gene to a VNTR lying 5' to the insulin gene, which affects the transcriptional activity for insulin production and influences the level of thymic expression of insulin during a critical period in the development of self-tolerance. Susceptible strains have a low level of *thymic* but a high level of *pancreatic* insulin mRNA compared with their resistant counterparts which could foster the insulin autoreac-

tivity often seen in the early stages of IDDM. (Yes, quite unexpectedly, it does look as though mRNA for a number of 'organ-specific' antigens can be detected in the fetal thymus!)

Unraveling complex polygenic conditions is a very tough assignment. If we may take murine SLE as archetypal, genetic analysis of the predisposition to disease is most compatible with a threshold liability model requiring additive, or epistatic, contributions of multiple susceptibility genes probably linked to different stages of disease pathogenesis (figure 19.5).

Hormonal influences in autoimmunity

There is a general trend for autoimmune disease to occur far more frequently in women than in men (figure 19.6) probably due, in essence, to differences in hormonal patterns. There is a suggestion that higher estrogen levels are found in patients and administration of male hormones to mice with SLE reduces the severity of disease. Pregnancy is often associated with amelioration of disease severity, particularly in rheumatoid arthritis (RA), and there is sometimes a striking relapse after giving birth, a time at which there are drastic changes in hormones such as prolactin, not forgetting the loss of the placenta. We should also note the frequent development of postpartum hypothyroidism in women with pre-existing thyroid autoimmunity.

In Chapter 11, we dwelt on the importance of the neuroendocrine immune feedback encompassing the cytokine–hypothalamic–pituitary–adrenal control circuit. Abnormalities in this feedback loop have now been revealed in several autoimmune disorders. Patients with mild RA have lower corticosteroid levels than normals or patients with

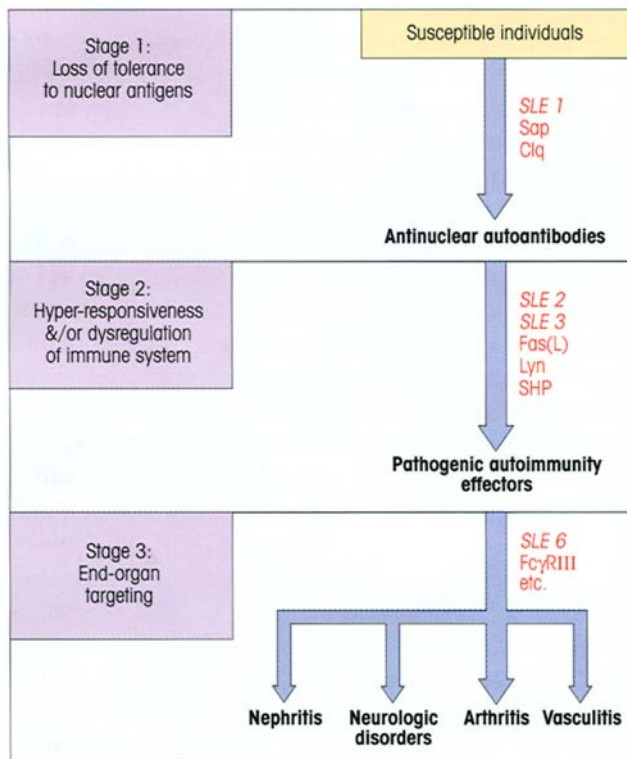


Figure 19.5. Possible stages in development of SLE in susceptible individuals. Genes or gene products are in red. A variety of genes are involved in end-organ targeting. Epistatic interactions between *SLE 1* and *3*, derived from the NZW strain, produce severe lupus when introduced together into an otherwise resistant B6 strain as a double congenic, even though neither alone causes severe disease. The NZW strain itself, which contains both genes, is resistant to disease because it has a genetic suppressor region, four of which have been identified in various murine lupus strains.

AUTOIMMUNE DISEASE	FEMALE : MALE INCIDENCE RATIO				
	2	4	6	8	10
SLE					
SCLERODERMA					
POLYMYOSITIS					
RHEUMATOID ARTHRITIS					
SJÖGREN'S SYNDROME					
AUTOIMMUNE THROMBOCYTOPENIA					
MYASTHENIA GRAVIS					
GRAVES' THYROTOXICOSIS					

Figure 19.6. Increased incidence of autoimmune disease in females.

osteoarthritis or osteomyelitis despite the presence of inflammation. Moreover, RA patients undergoing surgery manifest grossly inadequate cortisol secretion in the face of high levels of plasma IL-1 and IL-6 (figure 19.7), a phenomenon now attributed to defective hypothalamic control. The OS chicken, several strains of lupus mice and the Lewis rat, which is abnormally susceptible to the induction of autoimmunity, all show blunted IL-1-induced corticosteroid responses. Both T- and B-cells from NOD mice survive for abnormally long periods in culture and their thymocytes are relatively resistant to corticosteroid-induced apoptosis. This would imply that the feedback cycle is not operating at the lymphocyte level and could cause dysregulated immune function; the ability of IL-1 in-

jections to delay the onset of diabetes would accord with this view.

Does the environment contribute?

Twin studies

Although the 50% concordance rate for the development of the autoimmune disease *insulin-dependent diabetes mellitus* (IDDM) in identical twins is considerably higher than that in dizygotic twins and suggests a strong genetic element, there is still 50% unaccounted for. This is not necessarily all due to environment, since although monozygotic twins have identical germ-line immunoglobulin and T-cell receptor (TCR) genes, the processes of diversification of receptors and of internal anti-idiotypic interactions are so complex that the resulting receptor repertoires will be extremely variable and unlikely to be identical. Nonetheless, a later study on concordance rates for IDDM in monozygotic twins gave the extraordinarily high figure of 70% if they were DR3/DR4 heterozygotes, but only 40% if they were not. Thus, in the same disease, the genetic element can be almost completely dominant or be a significant but minor factor in determining the outcome. As we turn to the nonorgan-specific diseases, such as SLE, we find an even lower genetic contribution with a concordance rate of only 23% in same-sex monozygotic twins, compared with 9% in same-sex dizygotic twins. There are also many examples where clinically unaffected relatives of patients with SLE have a higher incidence of nuclear autoantibodies if they are household contacts than if they live apart from the proband. However, within a given home, the spouse is less likely to develop autoantibodies than blood relatives. Summing up, in some disorders the major factors are genetic, whereas in others environmental influences seem to dominate.

Nonmicrobial factors

What environmental agents can we identify? Diet could be one—fish oils containing long-chain, highly polyunsaturated ω -3 fatty acids are reputed to be beneficial for patients with RA; someone must know whether rheumatologists in Greenland are underworked! Sunshine is an undisputed trigger of the skin lesions in SLE. Exposure to organic solvents can initiate the basement membrane autoimmunity which results in Goodpasture's syndrome—witness the high incidence of this disease in HLA-DR2 individuals who work in dry-cleaning shops or siphon petrol from other people's petrol tanks. A more contrived situation

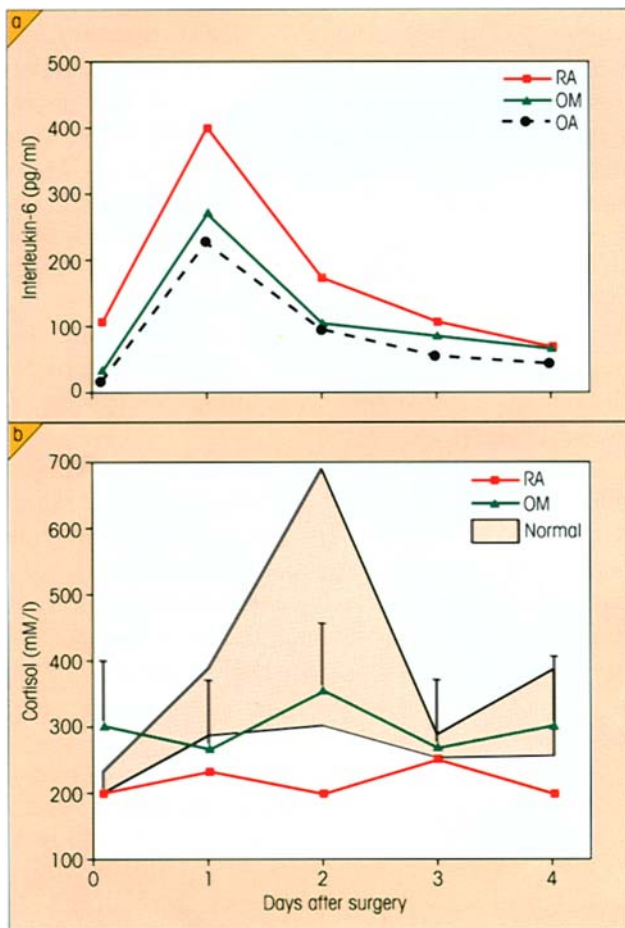


Figure 19.7. Failure of feedback control of cortisol production in rheumatoid arthritis (RA). After surgery (a) RA patients have even higher levels of plasma IL-6 than osteoarthritis (OA) and osteomyelitis (OM) controls. Nonetheless, (b) they have profoundly deficient production of cortisol which is evidence of faulty feedback control. (Data kindly provided by Professor G. Panayi from Chikanza I.C. *et al.* (1992) *Defective hypothalamic response to immune and inflammatory stimuli in patients with rheumatoid arthritis. Arthritis and Rheumatism* 35, 1281, with permission from the publishers.)

is the production of a similar disease in Brown Norway rats by the injection of mercuric chloride, but it makes its point, and there are several drug-induced diseases such as SLE, myasthenia gravis, autoimmune hemolytic anemia, and so on.

Microbes

Of course everyone's favorite environmental agent has to be an infectious microorganism and we do have some clear-cut examples of autoimmune disease following infection, usually in genetically predisposed individuals: acute rheumatic fever follows group A streptococcal pharyngitis in 2–3% of patients with a hereditary susceptibility and B3 coxsackie virus produces autoimmune myositis in certain mouse strains.

In most cases of human chronic autoimmune disease, the problem is the long latency period which makes it difficult to track down the initiating event (cf. figure 19.3) and, secondly, viable organisms usually cannot be isolated from the affected tissues.

Some groups are focusing down on **very slow-growing forms of mycobacteria** which are exceedingly difficult to culture but appear to be associated with various hypersensitivity states. *M. paratuberculosis* is linked with Jöhne's disease, a chronic granulomatous intestinal infection of cattle similar to Crohn's disease. Genetic probes have identified mycobacterial sequences in sarcoidosis. The granulomatous Takayasu arthritis is often accompanied by very powerful responses to TB. Whipple's disease associated with arthralgias and skin changes evolves so slowly that it can often take 10 years for the diagnosis to be made and, although organisms can be identified in macrophages at electron microscopy level, none can so far be cultivated. There has been a breakthrough in the HLA-B27-related **reactive arthritis** provoked by infection with *Chlamydia*, *Yersinia* or *Salmonella*, in that T-cell responses to bacterial fragments or perhaps to cross-reacting self-epitopes present in affected joints can now be demonstrated years after the primary infection. These studies and the ability of EB virus and *Chlamydia* DNA probes to hybridize to a not insignificant proportion of synovial tissues from rheumatoid arthritis patients raises important questions regarding the cellular localization and molecular status of the 'microbial' nucleic acid in these cells.

Given the possibility that we may ultimately identify persistent microorganisms in some disorders, we then have to re-examine the question of whether the hypersensitivity lesions are driven by the microbe or by self-antigen, i.e. are we dealing with a microbially triggered autoimmune disease or autoimmune phe-

nomena secondarily superimposed on an underlying microbial hypersensitivity? Perhaps both circumstances occur. Certainly we know that cross-reactions with microbial components can initiate autoimmunity (see p. 412), and recently it has been shown that infection with the helminth *Nippostrongylus brasiliensis* can break tolerance to an unrelated staphylococcal superantigen; perhaps infections of this nature can stimulate such a welter of cytokines as to nonplus and maybe activate anergic, potentially autoreactive T-cells. They can also act as superantigens in bringing about the polyclonal stimulation of certain TCR V β families. Further complexity is injected by the knowledge that environmental microbes may sometimes **protect** against spontaneous autoimmune disease; the incidence of diabetes is greatly increased if NOD mice are kept in specific pathogen-free conditions, while Sendai virus inhibits the development of arthritis in the MRL/*lpr* strain. The extraordinary variation in incidence of diabetes in NOD colonies bred in a wide variety of different animal houses (figure 19.8) testifies to the dramatic influence of environmental flora on the expression of autoimmune disease.

AUTOREACTIVITY COMES NATURALLY

Tolerance mechanisms do not destroy all self-reactive lymphocytes. Processing of an autoantigen will lead to certain (dominant) peptides being preferentially expressed on *antigen-presenting cells* (APCs) while others (cryptic) only appear in the MHC groove in very low concentrations which, although capable of expanding their cognate T-cells in the context of thymic positive selection (cf. p. 228), may nonetheless fail to provide a sufficiently powerful signal for negative selection of these cells. As a consequence, autoreactive T-cells specific for **cryptic epitopes** will survive in the repertoire which will therefore be biased towards weak self-reactivity.

Because conventional B-2 cells are less susceptible than T-cells to tolerization by low concentrations of circulating autoantigens such as thyroglobulin (figure 12.16), autoreactive B-cells specific for such antigens will circulate albeit unaccompanied by their cognate helper T-cells (figure 12.17). The reader will also recall the B-1 cell population, which starts off early in life by forming a network connected by germ-line idiotypes. The cells are stimulated, presumably by T-independent type 2 idiotype interactions, to produce so-called '**natural antibodies**', a term applied to those serum antibodies thought to be present before external antigen challenge and therefore arising independently of conventional antigen stimulation. These are